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Investigating Consumer Perception and Speculative AI Labels for Creative AI Usage In Media

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The growing importance of Creative AI and its uncertain impact on people necessitates more research. Current studies mainly focus on technical aspects, neglecting consumer perspectives. This study seeks to explore consumer perceptions of Creative AI in consumer media and introduces speculative labeling as a potential solution to empower consumer choice. Through Content Analysis of 32 Creative AI tools and a survey of 40 respondents, insights on AI's value, potential job displacement, and the need for AI usage disclosure emerged. These findings informed a workshop with 7 participants using the newly proposed N-Speculation method and uncovered consumers' bias against AI but indifference toward creators. Factors like cost, quality, and the creator (AI or Human) affect purchase decisions. Consumers favor labeling with clear pictograms and detailed AI usage information. These findings advance our understanding of consumer preferences and highlight the need for more consumer-centric Creative AI research. This research underscores the importance of education, addressing bias, and supporting the growth of Creative AI for all stakeholders.

SAMMANFATTNING

Den ökande betydelsen av Kreativ AI och dess osäkra inverkan på människor kräver ytterligare forskning. Nuvarande studier fokuserar främst på tekniska aspekter och försummar konsumentperspektivet. Denna studie syftar till att utforska konsumentens uppfattning av Kreativ AI i konsumentmedier och presenterar spekulativa märkningar som en potentiell lösning för att stärka konsumentens val. Genom innehållsanalys av 32 Kreativ AI-verktyg och en enkät bland 40 respondenter framkom insikter om Kreativ AI:s värde, potentiell arbetsdislokation och behovet av information om AI-användning. Dessa resultat utgjorde grunden för en workshop med 7 deltagare som använde den nyligen föreslagna N-Speculation-metoden och avslöjade konsumentens fördomar mot AI men likgiltighet gentemot skapare. Faktorer som kostnad, kvalitet och skapare (AI eller människa) påverkar köpbesluten. Konsumenten föredrar märkningar med tydliga piktogram och detaljerad information om AI-användningen. Dessa resultat fördjupar vår förståelse av konsumentpreferenser och understryker behovet av mer konsumentcentrerad forskning om Kreativ AI. Denna forskning betonar betydelsen av utbildning, att hantera fördomar och att stödja tillväxten av Kreativ AI för alla intressenter.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing** → **Empirical studies in HCI**; **HCI theory, concepts and models**.

Keywords: Creative AI, Speculative Design Thinking, Consumer Perception, Consumer Choice, Consumer Labeling, Near-future Speculation

Nyckelord: Kreativ AI, Spekulativt Designtänkande, Konsumentuppfattning, Konsumentval, Konsumentmärkning, Spekulation om nära framtiden

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1 INTRODUCTION

There are a few different positions on the spectrum of how Creative AI v. human creatives will play out: On the more technocentric end of the spectrum AI takes over and creativity is no longer an activity exclusive to humanity; on the more anthropocentric end, Creative AI is seen as just a fad, creativity is innately human [59], and any attempts to sever such a bond is a foolish endeavor[1]. However, more realistically, the future of creativity and AI lies somewhere in between: AI becomes the next big tool that enhances human creativity through disruptive technological means, where AI acts as more of a tool or colleague for co-creation rather than a full-out replacement of their human alternative. Early examples of this can already be seen in a multitude of areas and content types where Creative AI has been used in Hollywood productions [74], high budget TV shows [42, 60], a Netflix short-film [6], magazine covers for Cosmopolitan [49] and The Economist [80], large scale marketing campaigns [13, 19], imitating the song style of well-known music artists [2, 52], reading the news in place of newscasters on live TV [65], published academic papers [58, 61, 67], and comic books [40] (to name a few). While these uses of Creative AI may either be assistive to human creators or have unmatured outputs while used in isolation, it is important to note that this is only the beginning of such a technology, or in other words, this is the least technologically capable Creative AI will ever be. While none of the above uses of Creative AI have had any substantial social backlash, the use of Creative AI is a mixed bag. For example, in February of 2023, Clarkesworld[2] had to stop accepting submissions due to a large influx of AI-created stories [3, 43]. Similarly, multiple online art community websites have completely banned AI-generated art from being hosted on their platforms after artists dissented against the presence of AI-created art [8, 87]; However, there has been no widely reported uproar among consumers of such media that has used Creative AI.

Further, despite the plethora of studies surrounding the more technical aspects of Creative AI, research has seldom focused on consumerist aspects of Creative AI. The few studies surrounding people (let alone consumers) are often focused on the broader studies of perception, creating art with AI, or the legal/ethical/moral or similar aspects. As seen above, there are already various areas where AI has been used in mainstream media, from web articles to high-budget TV shows. This means that consumers are now going to be thrust into an era of products that they would spend their money on without knowing or even understanding the implications they might bring. Similar to consumers choosing organic food, produce that treats farmers ethically, or high energy efficiency electronics, consumers are often big decision makers that can move and demand that companies follow policies that are in line with their values (such as eco-friendly or pro-LGBTQ). However, it is currently unknown what might these values be, or in other words, *what do consumers think about Creative AI?* However, simply knowing these perceptions is only half as good if we do not know what to do with this data and how it could be used to positively affect the stakeholders targeted by this study. Thus, this thesis seeks to be a preliminary study on the perception of Creative AI on consumers and, using those understandings, aims to provide a route towards furthering consumer choice and education through AI labels.

This thesis attempts to fulfill the above research goals by first reviewing the relevant literature. The background section sets up definitions of important keywords: **Creativity**, **Artificial Intelligence**, **Creative AI**, and **Consumers**, and also reviews the existing literature surrounding consumers within the context of Creative AI. As this thesis looks to AI labels as a possible solution, the background section also investigates consumer labeling more broadly and uses those generalized insights to establish AI labels as an appropriate solution and inform their designs used during later methods of data collection. Three methods were followed to collect data and validate hypotheses; A content analysis of 32 widely available Creative AI tools was performed to understand the capabilities and relevant properties of existing

[1]The terms "dystopia" and "utopia" were initially considered. However, the use of these terms has been rescinded as they view Creative AI from a lens of extreme desirability/undesirability. This paper seeks to serve as, and recommends future studies to be, an unbiased work of academia aimed at furthering research rather than tilting the scales of acceptance.
[2]A science-fiction magazine that publishes short stories. Anyone can submit stories through their website and receive payments for accepted submissions.

AI tools. This content analysis provided insights into how AI could change the creative landscape, such as establishing if Creative AI can only be assistive or full replacements or replace only some jobs/tasks usually allocated to humans. The content analysis was used to design a survey questionnaire that was used to collect information about consumer perceptions. General consumer perceptions and attitudes were established and revealed a mostly confused yet curious population. Further, the survey also established the usage of labels (or declarations in general) for media that uses Creative AI (both digital and physical) as not only a viable but an essential request by consumers. In the second phase of this study, an N-Speculative workshop will be used (1) to validate the perceptions and attitudes obtained during the survey and obtain more qualitative data, and (2) to generate label designs through co-speculation to understand what elements consumers would like to find in such label designs. The data obtained from the workshop was analyzed using thematic analysis (for the discussions) and content analysis (for the label designs). The workshop provided further understanding (and validation) of the survey data and dove deeper into the fears, indifference, and curiosities of the participants; the co-speculation activity provided label designs that showed that above all else, consumers want pictograms for easy understanding, and want to be provided information on how Creative AI was used, to understand how the media consumed would align with their [still developing] morals and personal ethics.

2 BACKGROUND

This section of the thesis reviews existing and relevant literature and sets up definitions that will set the scope and delimitations of this study. A review of the background literature shows that most studies currently are focused on the technical aspects of Creative AI and not so much on how it affects people. The background section also covers the two important definitions of this paper: Creative AI and Consumer. These definitions are not built out of thin air but are derived from logically assembling ideas and definitions from various literature. This section also introduces the relevant literature on speculative design that will be used to establish the newly proposed method of N-Speculation in the next section (sec. 3).

2.1 Definitions

Across scientific literature, there exists vast disputes and postulations of definitions. Such semantic dissonance often leads to disagreements, not over the actual subject matter, but over misunderstandings [53]. Thus, key terms are defined prior to the discussion of the subject matter.

2.1.1 Artificial Intelligence. There exist a plethora of definitions for what constitutes AI [25, 47]. The most widely accepted (or recognized) definition of these is often attributed to Marvin Minsky[3] [23]:

> *The performance of tasks, which, if performed by a human, would be deemed to require intelligence.*

This definition is chosen as the basis for defining AI in this project due to (i) it's non-reliance on the technical aspects of AI, (ii) being broadly accommodating of various ideas, (iii) wide acceptance in the academic community [23], (iv) it's suitable congruence with Creative AI, and (v) it's congruence with the definition of creativity used in this project (2.1.2).

2.1.2 Creativity. Just as with AI, the definition of creativity has eluded a general consensus, altering its perspective to suit the relevant study [85]. Taking a similar approach to locating a definition of AI, the most common definitions of creativity and their respective uses were studied [12, 85]. From these, the most widely accepted and suitable definition [15] was identified:

[3]Though this definition is widely attributed to Marvin Minsky, the source of this definition is unclear [86].

> *Creativity is the ability to come up with ideas or artifacts that are new, surprising, and valuable. 'Ideas' here include concepts, poems, musical compositions, [...] and so on. 'Artefacts' include paintings, sculptures, steam engines, [...] – and many other things you can name.*

This definition is highly suitable due to its characteristics of (i) not specifying a particular creator *viz.* the creator does not have to be human, (ii) not imposing strict criteria on who or what is creative, and (iii) approaching creativity from the angle of ideas and products (*cf.* "Artefacts"), i.e., outputs. These outputs can also more broadly include the outputs of generative AI such as AI art, AI generated textual content, AI video scenes, etc. Further, this definition also provides appropriate criteria for what constitutes creativity: new, surprising, and valuable; "New" here can seem to mean anything that was not previously created, "Surprising" here means that the artifact is in some way surprising, for example, due to the way it was created, how it was created, or due to an abstract and non-discernible reason; "Valuable" here means that the artifact has some value. This "value" can be financial, sentimental, culturally important, or any other such value.

2.1.3 Creative AI. Using the definitions of AI and Creativity listed in sections 2.1.1 and 2.1.2 respectively, we may appropriately splice them together to *generate* a usable definition of Creative AI. Looking at the definition of AI, the word "tasks" can conveniently be assumed to include (as a subset) "creative tasks" and replace intelligence with "Creativity"[4] to better suit the needs of this project. Now, the keyword "Creative" can be defined using the selected definition of creativity, and organically arrive at a definition of Creative AI:

> *The performance of tasks to generate outputs which, if performed by a human, would be deemed to require Creativity[5], where Creativity is defined as the ability to come up with or generate, ideas or artifacts that are new, surprising, and valuable e.g. concepts, poems, musical compositions, art, film, or computer code.*

2.1.4 Who Is A Consumer? Just as with the previous definitions, this core term also requires some definitions to be set up. Unlike the above terms though, there does not appear to be much discussion on the attempting to define what *consumer* means or encompasses. In a dictionary, "consumer" is defined simply as *"one that consumes"* [57]. A similar sentiment is reflected in the literature [88] where the lack of definition (s) often lands on authors settling for a general definition based on the wider literature [36] or one that is specific to the respective paper or study. This study takes the latter approach as the semantics of the word are outside the study scope, while the need for a definition is necessary for providing the required context to successfully study the subject matter and successfully interpret the results. Thus, in this paper consumer is defined as:

> *An individual who consumes media content (simply media hereon) by paying for media through a monetary-based transaction (money for media), including but not limited to, purchasing physical/digital media through a one-time purchase, paying a monthly subscription, or using an ad-supported service where the individual is presented with an advertisement that temporarily halts consumption.*

From the above definition, it can be questioned what "consumption" refers to. Hence, the opportunity is taken to also define consumption as:

[4]While it is still widely debated if creativity and intelligence are in any way correlated [12, 26], such an exploration is outside the scope and requirements of this project. Additionally, since *"Artificial"* and *"Intelligence"* are not being discussed in isolation but rather as a singular term, creativity and intelligence can be interchanged.

[5]Creativity (and by extension Creative, Creation) has been used with an uppercase "C" to indicate that this refers to the definition provided in the paper and differentiate it from the dictionary noun which has also been used in relevant places. The use of the uppercase "C" also subscribes to the notation of artificial intelligence as AI (in uppercase letters).

> *The act of experiencing media either, actively or passively, including but not limited to, listening to vinyl actively, listening to music in the background while performing a different main task, streaming a movie, reading a book, playing a video game, reading an article, etc.*

2.1.5 Critical Reflection On Definitions' Impact On Study Framing. While the various advantages of these definitions are provided in the terms' respective sections, this section provides some critical perspectives and limitations on the use of these definitions, to complement and complete the construction of these terms. Similar to any definitions found across literature, these definitions will also be subject to interpretation, introspection, and criticism. For example, it might be argued that using Minsky's definition of AI might be too simplistic since there is no measurement of what constitutes intelligence, which leads to a situation where the study does not take a clear stance on the nature of intelligence; it may then be argued by some that, by their own definitions of intelligence, Creative AI cannot be intelligent (or creative). Beyond an acknowledgement, this study is not concerned in exploring of these potential drawbacks since the provided definitions provide enough context for the study activities to be performed and the results to be interpreted. However, these arguments are welcomed as they advance not just the Creative AI research, but also adjoining areas such as creativity research and semantics.

2.2 Consumerism And Creative AI

Creative AI has been around for almost a decade [18], though it entered into the mainstream public discourse only around 2022 [31] and sparked major discussions about art [7], creativity [7, 45], and its impact on creatives [7, 24]. While these discussions are far from conclusive, the academic literature has been lacking in the context of the impact that such technology creates on people. Outside of technical studies, most *people-focused* literature centers around perception of AI generated art [29, 34, 39, 59, 62, 69], the enhancement of creative efforts using AI [5, 41, 48, 50, 51], or explorations of legal areas such as copyright or ownership [32, 54]. The sparsity of literature that persists indicates that efforts have seldom taken place to understand the direct impact of this emerging technology on the various stakeholders. While some stakeholders have received some research recognition, for example, Twitter users [38], various creative professionals [51, 62] and non-creative professionals [30, 89], research around some stakeholders, such as consumers, is largely absent. Consumers are an extremely vital part of not just free markets, but also of the same creative industries that are being affected by creative AI: consumer demand and choice dictate, the price, quality, quantity, type, etc. of various creative products and endeavors. Given the lack of research surrounding Creative AI and consumerism, this study attempts to provide preliminary insights on the subject matter. Further, with "consumerism" being a wide field of study, focus is placed on consumer choice, specifically an explorative overview of the attitudes of consumers on the use of Creative AI in the products or content they purchase or consume.

2.3 AI And The Need For Informed Consumer Choice

Current research shows a clear perception bias against AI-generated content: people are more likely to favor works created by humans [34, 69]. However, research also shows that humans cannot tell apart AI-generated content from human-made content [29, 34, 73]. Based on this dichotomy it is fair to assume that consumers might want to know if AI was used in the products they purchased. Providing consumers with information that enables them to make choices is not new. Labeling has been chosen for this project as labels, such as eco-labeling, have shown to significantly affect consumer choice [14, 78]. Nutrition labels, eco-labels, ingredient lists, country of origin, parental ratings, and region numbers for DVDs, are just some of the kinds of information provided to consumers for them to make informed purchase decisions. Similarly, "AI labels" would provide consumers a say in choosing to purchase AI created works since AI has similarly massive effects, such as for example, affecting livelihoods [30]. In fact, 2023 the EU commission

has also floated to tech companies the idea of labeling AI-produced content [46]. It is fair to hypothesize that consumers might also want to be informed about AI usage. Once such speculation is accepted, questions such as the following arise: Would consumers have to be provided with information on whether Creative AI tools were used in the creation of a painting, a blog post, a YouTube video, or even clothing? What information, if any, would they like to be provided with? It can be seen certain organizations have already begun to take an active step to provide this kind of information; Wired Magazine has declared that it will disclose the use of AI-generated tools [64], CNET has created the "AI Assisted" declaration [37], and DeviantArt tags artwork made using generative AI [77]. While no study exists yet to determine if people would like to be informed about the use of Creative AI technology, the subject matter it is certainly worth exploring.

2.4 Exploring Consumer Choice Information Enhancement Through Speculative Workshops

Speculative design aims to utilize current understandings to imagine preferable futures [27]. Speculative design has often been (rightly) criticized for its lack of public/stakeholder inclusion and producing showpieces meant to be exhibited [44]. However, this speculative design approach can be a useful to study and understand future possibilities when used as a tool for prediction rather than imagination and serves as a vital method for exploration, examination, or a reflection of these future possibilities. The power, capability, and future prominence of AI have been established throughout the background section; however, this study bases itself on scarce literature, and as a result, needs to obtain data to make these predictions as accurate as possible. Unlike other speculative studies, this study does not aim to be critical or produce thought experiments [11], but rather (preliminarily) provide a seamlessly continuous *'current'* and *'future'* picture. With speculative design on one end and more traditional research-driven design on the other, there does not appear to be a design research method that serves a more *temporally immediate* purpose, *viz.* covering only probable futures rather than a range of probable to possible futures (ref. fig. 1)[21]. Thus, this study proposes a new design research method called Near-future Speculation (or N-Speculation) that builds on the existing foundational idea of speculative design [27] but is less critical, less aesthetic focused, research-oriented, employs the participatory aspect of co-speculation, and is more temporally immediate. Further, this speculative method seeks to anchor the outcomes in a more near and probable future, thereby aiming to bridge the gap between traditional design research and speculative design research by providing the wider scientific body with literature that can be used as traditional scientific reference and as design inspiration. The need for this new type of speculative design method arises because traditional speculative design methods are not very suited for scientific study, while traditional design research is too grounded in existing theory and data. N-Speculation thus provides a middle ground where speculative design allows users the freedom to venture beyond the problem-solution model of traditional design research, while also anchoring them to theory and data rather than the fantastical lands created by traditional speculation; this combination allows researchers the flexibility to test concepts that are not fully understood or well-studied, such as in this case consumers choice empowerment through AI labels. Through N-Speculation, researchers no longer need to wait for concrete scientific studies to test concepts and hypotheses that *feel like they could work* but there exists a lack of existing literature.

While the type of approach has been critically evaluated to be N-Speculation design, a delivery method must also be chosen. Workshops as the method of choice for prospective and 'unstructured' domains have been well documented and studied, including their use in participatory design [91]. Consumers are central to the results and future use of this study and are also the core stakeholders to any applications of this research; As a result, a collaborative workshop must be conducted where, though the researcher is in control, participants 'drive' the workshop [20].

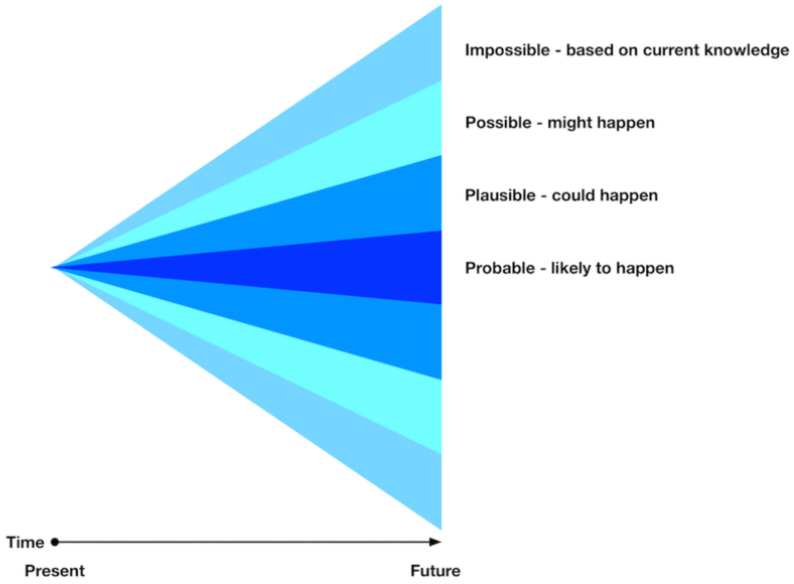


Figure 1: Futures cone indicating a a range of potential futures [21].

3 METHODOLOGY

This section will describe the methods in detail. Three methods were chosen, each serving as a chaining link between each other: **Content Analysis**, **Survey**, and **N-Speculative Design Workshop**. First, a brief content analysis of some widely available Creative AI tools will be conducted; this is done to understand the current capabilities of these AI tools, such as if they could replace any jobs entirely/partially. The information obtained from this will also aid in designing the types of labels that will be used in the survey and the design workshop. Next, a survey will be conducted to understand the perceptions of consumers on various aspects of Creative AI. The data from this will also be used to inform the creation of the speculative discussion prompts that will be used during phase 1 of the N-Speculative design workshop, and the questionnaires prior to the start and at the end of the workshop. Finally, the workshop will serve to collect data on consumer perceptions (phase 1) that can be cross-checked with the survey data and provide insights into the design of speculative consumer labels for AI usage.

3.1 Content Analysis And Survey Informing Label And Workshop Design

The survey and content analysis both provided insights into two different perspectives on the core issues of the study, consumers, and AI tools (respectively). The data acquired from these two methods informed the design of the labels and the speculative discussion prompts to be used in the workshop. There are two sides to this study: AI and People. To look at each side, first, a content analysis of some well-known Creative AI tools and technology is carried out to understand their present capabilities and their potential future uses; then a survey was conducted based on these potential future capabilities to get a picture of how humans would react to these and understand how they perceived the various aspects of these technologies and their *implied* consumer use.

3.1.1 Content Analysis. A content analysis was performed due to its flexibility in interpreting largely unstructured and varied qualitative data. The following criteria were applied to select which tools would be analyzed:

(1) Must be available to the general public either as a deployed tool, or compliable source code. Thus, tools that are, for example, available only to enterprise customers are not included in the analysis.
(2) Must have been publicly and openly available (incl. no waitlists) before August 1, 2023. The excludes tools that are "coming soon", "closed beta", "invite only", or "early access" as these tools are still in the very early stages of development and can be buggy, leaving their capabilities to become improperly evaluated.
(3) Tools must utilize Creative AI, as defined in 2.1.1. This means that AI tools such as Photoshop's content-aware fill, automation tools, or tools that utilize more generic AI do not qualify.
(4) "Embedded versions" of the same tool are counted as one tool. For example, Adobe Firefly has been integrated into Photoshop, so Photoshop is not counted as a separate tool. However, this does not include implementations of the underlying technology such as OpenAI's GPT models.

Understanding the capabilities of these tools will also aid in understanding how AI could change the creative landscape; for example, if Creative AI can only be assistive today rather than a full human replacement, or which media (books, music, video games, etc.) can be impacted by Creative AI in the near future and those that cannot. Such data from the content analysis informed the questions of the survey.

3.1.2 Survey Questionnaire. An online survey was conducted using an online survey tool[6] to understand how AI Usage labels in Consumer Items influenced consumers, and the importance that they place on various parameters. The survey consists of 14 questions[7], 8 primary questions aimed at addressing the survey goals, and 6 additional demographic questions to better inform the survey findings due to use of convenience sampling. The survey included a question where the three viable combinations discovered from the Content Analysis (ref. section 3.1.1) were included as types of declarations that consumers might want to see (ref. table 8 in Appendix C.2):

- **100% Human Creation and AI-Free:** This option indicates that no AI was used. Though presented as different options, they are effectively two sides of the same coin.
- **AI Assisted:** AI has in some way helped in the creative process but not fully produced it.
- **AI Produced:** AI has fully produced the Creation.

Everybody can be a consumer of some creation that has, is, or going to, use Creative AI technology. This may include (but is not limited to) books, music, art, film, etc. Using our definition of Creativity, we can say that any work that satisfies that definition can be considered "Creative Work". Hence the population of such a survey will include any person who can consume Creative work, where consumption is in a manner appropriate to that particular work (ex. Watching a film is appropriate but listening to a painting may not be). Since the population of this survey is non-specific, non-random convenience sampling is both suitable and advantageous for being quick and open to a wide array of respondents [35]. The survey was distributed primarily via social media channels (LinkedIn, Instagram), online forums, and personal IM platforms.

3.2 N-Speculation Design Workshop

A two-hour N-Speculation Design workshop was conducted to (1) determine current attitudes and perceptions of consumers towards Creative AI, and (2) Understand what aspects of AI Use consumers would like to be informed about through labels. Participants did not need to meet any specific criteria as anyone can be a consumer. Participants' demographics (ref. table 2) were collected to better inform the results of the workshops. The workshop consisted of 4 phases:

(1) **Pre-Workshop Quantitative Perception Measurement:** Participants were provided a 9-item questionnaire containing multiple choice questions that evaluated various aspects of their perception or attitude towards Creative AI. This data is

[6]The tool used was QuestionPro.
[7]Survey respondents reported not understanding what Question 7 meant, thus this data was discarded.

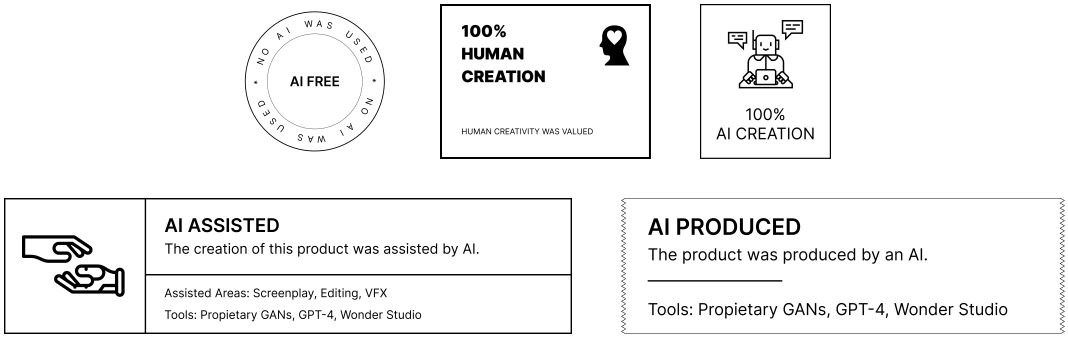


Figure 2: Pre-Designed AI labels indicating the various states of 'AI usage'.

collected to evaluate and compare these various aspects before and after receiving more information on the subject matter to see if, upon learning more, user perceptions and attitudes are altered. The questionnaire can be found in table 11 in Appendix D.3.

(2) **Familiarization through discussions:** In this phase participants discussed their feelings, thoughts, attitudes, and perceptions toward various aspects of Creative AI. Participants were given a speculative discussion prompt (ref. table 10 in Appendix D.1) each consisting of a unique scenario that would trigger various feelings, ethical quandaries, thought experiments, etc. Prompts were changed when discussions died down. This phase got participants acquainted with as many viewpoints as possible to (i) broaden their informed decision-making for phase 3, and (ii) check if their perceptions had changed during phase 4.

(3) **Co-Speculative Label Design:** During this phase participants were provided with a worksheet and the following speculative scenario:

> "In the year 2035 Generative[8] AI use has become mainstream in all forms of media production. This has led to some debate in society on consumer choice. You are all designers of an organization that is tasked with designing a label that seeks to promote consumer choice, in a manner similar to how eco-labels empowered consumers to choose more eco-friendly alternatives in the 2020s."

The above scenario is used by participants to design labels for a future where the use of Creative AI requires to be declared in some way. Participants were provided with 15 minutes to design their labels. Post this, participants would discuss in the larger group the thought process behind their label designs, and either ask/answer questions, or provide feedback to other participants. At the end of these discussions, participants place their worksheets at the center of the table and use the green voting dots to vote for the design they like the most (one vote per participant).

(4) **Post-Workshop Quantitative Perception Measurement:** Participants are provided with an identical questionnaire to the pre-workshop session to measure any changes (if at all) in their perception of the various aspects measured in phase 1.

3.2.1 Label Design. Using the results from the Content Analysis and Survey Questionnaires, five labels are designed to be used on mockups of physical and digital media. The types of these labels were determined from the content analysis of the 32 AI tools. The messaging for each of the labels was based on the data from the survey, such as for example, the label "100% Human Creation" contained the tagline "Human creativity was valued" as a reflection of the sentiments echoed in the qualitative survey data. All labels were designed to primarily showcase the variety in aesthetics, information design, content structure, and target applications. The icons and layout used are arbitrary and only serve the purpose of presenting the *label concepts* clearly to the workshop participant. Color was purposefully avoided to reduce the number of variables being studied.

[8]Generative AI was used here as it was more readily understood by participants in comparison to the term "Creative", though the results were still in line with the study definition of Creative AI.

Label	Short Description	Long Description
AI Free	No AI was used.	The label is circular (like a seal) with the words "AI Free" written in the center and the text "No AI was used" in the boundaries. No icons were used in this label. This label is the same as the next one below, though informing the consumer from a technological standpoint
100% Human Creation	No AI was used.	The label is a box with the words "100% Human Creation" written in the top left with the icon of a human head with a heart in it. A tagline at the very bottom of the label reads "Human Creativity was valued" to indicate that this label is specifically geared towards promoting the ethical production of creative items and is against the replacement of humans by AI. This label is the same as the one above, though informing the consumer from a human standpoint.
100% AI Creation	No Humans Involved	This label is similar to a stamp in containment and consists of a large robot icon in the top two-thirds of the label with the words "100% AI Creation" in the lower third. This label is the opposite of the previous two above.
AI Produced	Only AI has created this product	The label is a box divided into two parts. The top two-thirds contain the title "AI Produced" and the descriptive tagline "The product was produced by an AI". The bottom third contains information on the AI "Tools" used. The label consists of no icons and has perforated edges simply to create aesthetic variety when presented to the workshop participants.
AI-Assisted	Both human beings and AI have created this product	The label consists of a box divided into three parts. A full-height and one-fifth-width box contains an icon of a human and AI hand one above the other to signify a *Human-AI connection*. The remaining width is divided into two almost equal parts. The top half consists of the label title "AI-Assisted" and a descriptive tagline: "The creation of this product was assisted by AI". The remaining bottom half is dynamic and changing depending on the application. It consists of two lines placed vertically, the first indicates the "Assisted Areas" to signify what part of the product utilized AI, and the bottom text lists the AI "Tools" used.

Table 1: Labels that were designed for the workshop and their descriptions.

P.ID	Age	Gender	Nationality	Job or Profession	Used Creative AI Tool
P1	DND[1]	Female	Sweden	Journalist[2]	Yes (ChatGPT)
P2	24	Male	India	PhD Student	Yes (Github CoPilot, ChatGPT, Mid Journey)
P3	24	Male	India	Student	Yes (ChatGPT, Quillbot)
P4	36	Female	Iran	UX Researcher	Yes (ChatGPT, Bing)
P5	27	Male	India	Student	Yes (no specifics provided)
P6	27	Male	India	Student	Yes (ChatGPT)
P7	36	F	Pakistan	Marketing Automation Specialist	Yes (ChatGPT, DALL·E)

[1] DND — Did Not Disclose

[2] This participant did not disclose their profession on the demographics sheet, but stated this during the discussion stage of the workshop.

Table 2: Participants Table

3.2.2 Analysis Of N-Speculative Workshop Data. The data collected from the workshop was analyzed using two methods:

Thematic Analysis. Qualitative data gathered during the discussion phases was analyzed using thematic analysis [16]. Here, the captured audio data was transcribed using Microsoft Word Online's built-in transcription feature. This text was then be checked against the recordings for any errors. The transcripts were then be imported into QDA Miner Lite coding software. The coding process took place in two phases; first broad coding was performed to simply highlight important or interesting parts of the data without worrying if they fit with other codes. After this, codes were amalgamated to better fit into more encompassing yet well-defined codes, which were then categorized into themes. The textual data was revisited to ensure that all the relevant data was coded and categorized.

Content analysis. Content analysis was performed on the labels designed by the participants. While this is less well-defined than thematic analysis, this analysis looked at which design elements were used, the reasonings behind them, the kind of information depicted, etc. to discover emerging themes.

4 RESULTS AND ANALYSIS

In this section, the results obtained from (1) Content analysis of various Creative AI tools (sec. 4.1.1 - 4.1.2), (2) Survey of consumer perception and attitudes towards Creative AI (sec. 4.1.3 - 4.1.6), and (3) Workshop containing discussions to assess consumer attitudes and behavior (sec. 4.2 - 4.2.4), co-speculative activity aimed at understanding

what consumers would like to see in AI labels (sec. 4.3), and the pre and post workshop perception comparison (sec. 4.4), is presented.

4.1 Content Analysis And Survey Informing Label And Workshop Designs

A content analysis of 32 Creative AI tools was performed. These tools were categorized according to the media they output, transparency, and pricing model. A list of these tools and their classifications can be found in table 5 in Appendix B. Post this, an online survey collected data from 40 respondents ($n = 40$). The demographic data of the participants can be found in table 6 in Appendix C.2. Questions 1-5 used a 5-point Likert scale (0-4) to measure the importance of various aspects. While referring to this data, negative terms are utilized for data reported as "Not at all important" and "Slightly important", positive terms are utilized for data reported as "Very important" and "Extremely important", and neutral/undecided terminologies are utilized for data reported as "Moderately important".

4.1.1 AI Tools Offer Capabilities In A Broad Variety Of Mediums Though In Varying Capacities. Analyzing the 32 chosen AI tools it can be seen that they are capable of generating images[9] ($n_I = 12$), code ($n_C = 5$), text ($n_T = 4$), speech audio ($n_S = 4$), music ($n_M = 3$), video ($n_V = 4$), and game assets ($n_G = 2$). Looking at the tools it can also be seen that most tools will be used in collaboration with human creatives, though some tools, particularly image generators, can be used standalone and entirely replace certain creative professions (such as junior illustrators); Of course not using any of these tools automatically results in the work being *100% Human Creation* or *AI-Free*.

4.1.2 Paid And Proprietary, Free And Open Source. Various pricing models are employed by various tools with the large majority ($n_{PPU+PTU+Fm} = 19$) charging for use in some manner. The most widely employed pricing model is the freemium pricing model ($n_{Fm} = 10$). For tools that cannot be used as a limited free version, most tools interestingly employed a Pay-Per-Use model instead of a Pay-To-Use model, where in the former, users are charged for the usage amount which may be measured by the number of *"compute units"*, time of usage, power usage, etc. This result might be used to infer that the cost of development or setting up the model/tool is negligible, thus potentially making wide scale use of Creative AI models a feasible alternative. From table 5 (Appendix B) it can also be seen that most tools are closed source ($n_{Cl} = 27$) with only a small minority of tools being open source ($n_O = 5$). Comparing the pricing model and transparency, it can be seen that all paid tools are closed source ($n_{PPU} \cup n_{PTU} \cup n_{Fm} \cap n_{Cl} = 19$). While this is bad for consumer transparency, it makes sense for the tools developers who would want to protect their market position, business, and the cost of development; however, there does not seem to be a reason why some free tools are closed source ($n_F \cap n_{Cl} = 6$). Information on how these tools operate is important (though not vital) to consumer awareness. On the flip side, the presence of free and open-source tools means that businesses can implement Creative AI tools in any manner they wish to see fit to their organization which, perhaps, might bring down costs further when compared to the already reduced cost of developing proprietary AI tools over using human creatives in the long term. Due to a combination of the unviability or business unsustainability of companies disclosing trade secrets of how their AI models operate, and how easy it is to implement these models at scale for low cost, the proposal of consumer labels for AI is a convenient compromise that allows companies to not disclose trade secrets while the consumers still receiving information that helps them make choices based on their personal choices and morals.

4.1.3 Consumers View Physical And Digital Media In The Same Light In The Context Of AI Usage. There does not appear to be much overall difference between the need for consumers to know AI use information in physical and digital media with a $\Delta\bar{x} = 0.13$ ($\bar{x}_{physical} = 2.8$, $\bar{x}_{digital} = 2.93$) slightly in favor of digital media. This lack of difference was also noted during the workshop though some participants noted that they would pay less for digital art

[9] Images here also refers to vectors, photographs, drawings, and similar outputs.

created by AI such as, for example, P_5: "If it's a real painting, you can see the texture. That I'll pay more [for]." Looking at the individual scores however, we can see that consumers would like to be informed about what AI tools were used for creation and if AI has either produced or assisted in those creations more so in digital media as when compared to physical media products. Interestingly when it came to finding out compensation information, on average consumers did not want to know this information for digital media; though the exact reason is not clear (discussed in more detail in 4.1.4). Further, looking at the average scores for each area investigated (Figure 4 and table 7 in Appendices C.3 and C.4 respectively) we can see that on average it is moderately important for consumers to know if the details of AI used in the creation of the media they consume/purchase. Overall, though there is some variation in what they would like to see and where they would like to see it, from figure 5 in Appendix C.5 it can also be seen that on average consumers would like to see labels in fairly equal measure on both digital and physical media.

4.1.4 Less Than Half Of Consumers Want To See Compensations For Loss Of Livelihoods. The survey data indicates that most respondents either cannot say or do not agree that livelihoods lost to AI should be compensated. In the survey, only 35% ($n_{Yes} = 14$) agreed that people replaced by Creative AI systems should be compensated. While only 3 of the 14 respondents provided a reason (and thus nothing significant can be concluded from it), all 3 responses talk about the ethics/morality of the situation:

(1) "A lot of AI is already producing materials based on humans' work."
(2) "People need jobs. Just something to set them up to find another job or provide training to reach new areas".
(3) "I don't think people should be replaced by AI at all to begin with unless it is for something that betters the planet & is overseen ethically by humans."

In contrast, 20% ($n_{No} = 8$ responded that they would not be in favor of such compensations. When asked why, one respondent compared humans being replaced by Creative AI systems to painters being replaced by the invention of photography, or postal works to email: "Technology will always be changing the job market, why is AI different here?". Another respondent, who interestingly reported to be a graphic designer and was well aware of the possibility of being replaced by AI, also disagreed with compensations stating: "... its our responsibility to find ways to be useful with advancements of technology ..." adding that market conditions will cchange,and people need to adapt. Another respondent had a rather blunt take about the validity of *impacted* jobs: "If the job can be easily replaced by a few algorithms, then did the job really need to exist? Sounds like a people cribbing about calculators stealing jobs". However, the vast majority of respondents (45% or $n_{IDK} = 18$) reported that they "don't know". Of these 18, 10 had provided reasons to back up their choices. These choices can be placed into 3 groups[10]:

(1) **Borderline Yes:** The singular response in this category indicates that though the respondent is in favor of compensations, they are unsure of how it would work.
"Carbon offsetting doesn't really work well, and I think here you also bring in the issue of proving you need to have lost livelihood, which is tricky. But we should not make it impossible for people to do certain professions that they could have done otherwise."
(2) **Borderline No:**, As opposed to the previous item, the responses from this group, indicate that the respondents might not be in favor of compensation. While one respondent in this category reported that Creative AI systems are currently incapable of replacing [a significant] number of people and thus there is no need to think of such compensations, all 4 other respondents state that changing market conditions are a fact of technological advancement and AI is simply the next advancement.
"I'm not sure how other people who get laid off due to similar circumstances would feel about only one genre of laid-off employees being compensated and why others are not".

[10]The group counts are: $n_{BorderlineYes} = 1, n_{BorderlineNo} = 5, n_{Justdon'tknow} = 4$

(3) **Just don't know:** The remaining 4 respondents stated the complex nature of this problem and their lack of understanding as the reasons for choosing this option, or in other words, they really "don't know".

"I am not really sure of how it should work really. Ai has a lot of potential to do tasks specifically those that involve a linear and repetitive nature (ex accounting, reading documents). It is justified by the employer as it saves team sizes and costs. While an entire field of work getting obsolete for people working in it is also not seems right."

Overall, there appears to be mostly negative outlook by consumers on livelihoods lost by AI, though there does not appear to be any conclusive evidence as to why.

4.1.5 Factual Transparency And Personal Connection To Media Stated As Primary Reasons For Requiring The Provision Of AI-Information. Question 6 asked respondents which specific content they would like to be informed about the use of Creative AI about; In combination with Q1 and Q2, we can get some interesting insights into future labeling.

Among the largest count were sources of information, *viz.* Scientific literature ($n_{scientificLit} = 27$) and News articles ($n_{newsArticles} = 23$). It could also be argued that YouTube Videos might also qualify as sources of information for a lot of people and hence also might explain why it ranks the 2nd highest ($n_{youtube} = 24$); this contrasts with less consequential items of Review Articles ($n_{reviewArticles} = 21$) and Blog Posts ($n_{blogPosts} = 19$) as sources of information. With the exception of Blog Posts, Comics, Physical Art Pieces, Software Products, and Video Games, for every other media category at least 50% of all respondents agreed that information on Creative AI use should be required (ref. table 9 and figure 6 in Appendix C.4). Interestingly video games scored the lowest with only 9 respondents suggesting that video games be labeled with Creative AI use information, though the reason for this is not clear. The subjective data (10 of the 15 participants that reported) to these questions reveals that there are two main reasons why respondents would like to see labels:

(1) **Transparency of factual reporting/sources:** The primary reason (reflected by both the count and subjective answers to Q6, and the workshop discussions) was to evaluate the credibility and potential biases of the content that they are reading. In addition, it appears that consumers are uncomfortable with the potential capabilities and power of AI.

"For anything non-fiction (news article, youtube, or scientific paper) it would be useful to know if video footage or photographs are real or not. For art/entertainment I don't care if its AI generated."

(2) **Personal connections:** The remaining 5 respondents stated some personal reason or connection with the media as a reason to be informed of the use of Creative AI.

"As a graphic designer each one of these things either i use directly in my work, or would be interesting enough to use myself for future opportunity..."

These patterns were also seen during the workshop discussion where participants stated that they would not be comfortable with Creative AI used in sensitive areas of writing such as (in this case) media coverage of the war in Ukraine (ref. 4.2.1).

4.1.6 Labeling. When it came to what kinds of labeling consumers would like to see, a significant number of respondents indicated that the type of label did not matter as long as a declaration was made ($n_{physical} = 13, (32.5\%)$ and $n_{digital} = 15, (35\%)$). On the flip side, a very small minority of respondents reported to not care about such labels ($n_{IDC(physical)} = 9, (22.5\%)$ and $n_{IDC(digital)} = 4, (10\%)$).

However, looking at the qualitative explanations, we can see that the reasons behind the respondents' choices differed (overall) between physical and digital media. 21 respondents provided qualitative explanations for their choices for

physical media, while 16 provided qualitative explanations for their choices for digital media. Of these 19 responses provide usable data for physical media, and 12 for digital media. Each of these responses can be grouped into categories (discussed below). Figure 6 and table 9 in Appendix C.4 provide the counts for each of the choices. While there were no discernible patterns between responses and choices made, taking a look at table 8 in Appendix C.4 it can be seen that while "AI Free" was the lowest picked labeling copy in both media types, when the option of "100% Human Creation" was provided. This could indicate that consumers value the "human spirit" higher than the lack of AI (evidenced by the discussion below under "The Human Aspect").

Reasons Respondents Wanted To See Labels On Physical Media. Participants responses can be grouped into 4 major reasons for wanting to see physical media labeled with AI-related declarations: **Monetary Factors**, **Curiosity**, **The Human Aspect**, and **Consumer Choice**. A majority of responses (7 of 19), indicated curiosity where respondents were more interested in finding out how the media was produced, what tools contributed to their making, or just a general interest in AI.

> *"I'm interested in AI, and seeing such labels would possibly make me more interested in product."*

4 of the 19 responses indicated that labeling was important as they would like to understand what happens to the human being replaced by AI. Their responses were thus either empathetic towards human workers and/or negatively directed towards corporations.

> *"It is important that people use their brains & hand to eye coordination skills. It is also important we have ethics around the use of AI. People are becoming more greedy & less empathetic, the over use of AI does not help this, People need jobs."*

3 out of 19 responses indicated that they would like labels to be provided as it would enable transparency and empower their decision-making as consumers while purchasing physical media.

> *"Creates a sense of trust and transparency about what I am being offered".*

Though only a singular respondent provided money as a factor for their choice, it is interesting to see that consumers might base their spending choices on whether a product was made by AI or a human being. This finding could perhaps be related to the handmade effect discussed in 4.2.3.

> *"If its AI made then I would expect it to be a bit cheaper".*

Reasons Respondents Wanted To See Labels On Digital Media. Participants responses can be grouped into 3 major reasons for wanting to see digital media labeled with AI-related declarations: **Curiosity**, **The Human Aspect**, and **Trustworthiness**. Similar to physical media above, curiosity was the most provided reason (5 of 12) with similar reasons of knowledge and interest.

> *"Interesting to see how much AI been used in digital content".*

While **The Human Aspect** is a category just as with physical media, the reasons, however, differ. With physical media the reasons were more empathetic towards the conditions of human workers, in digital media the responses indicated empathy towards the work (media) instead, *viz.* towards the *soul* of the creation. A similar reason was heard during the workshop discussions discussed in 4.2.3.

> *"Depends on the content being consumed but imo, in most cases the involvement of a human imparts certain value to the final content through their experiences and stuff... which is not necessarily true for ai stuff which can only try and emulate it. The additional context of who the creator is, holds value to me."*
>
> *"I want to know if content is original human thought or a compilation of data that may have questionable sources".*

Reasons Respondents Did Not Care About Such Labels. It is also of interest to look at the reasons provided by respondents (1 for digital, 4 for physical) for why they did not care about such labels. One respondent said that they were against "any kind of mandatory labeling system" for both digital and physical media. They stated that: "Existing certifications like UL and FCC already pose a large financial barrier for small companies and independent developers...Adding additional mandators certifications only makes that problem worse."; However, the same respondent was "in favor of regulations that define what labels like AI-Free and AI-Produced actually mean so that the labels can be useful to consumers if the manufacturer chooses to put them on a product." Taken at face value then, this statement can be interpreted as the respondent in favor of labeling from a consumer choice point of view, but not a regulations point of view, *viz.* optional labeling.

While no other reasons were provided against labeling of digital media, physical media had 3 more distinct comments (rather than reasons). One respondent reported that they "...don't really care who created the physical product." "As long as it is not a product whose manufacturing requires quality checks/regulation (medicines, building materials, etc)...". This response is interesting as the respondent seems to be thinking about Creative AI used in more futuristic contexts such as medicine (though however, it is also possible that they have considered Creative AI and AI more generally as the same). Another respondent commented that though they were unsure about how to feel about AI they felt that "...it would be nice if it is disclosed when ai is being used...due to their potential". While these two respondents appear to be at least a little bit open to accepting the usefulness of labeling, one respondent outright stated that they "do not think it is that important as this era is extremely digital" indicating that some consumers might just see AI as an expansion of technological progress.

4.2 Results From N-Speculation Workshop Discussions

Two workshop sessions were conducted, one with 3 participants and another with 4. Participant[11] demographics are provided in table 2. A reflexive thematic analysis was carried out and the resulting codes and themes generated are provided in table 3. The workshop consisted of two parts: (1) A discussion phase where participants discussed their thoughts and feelings surrounding speculative prompts which investigated various aspects of the use of Creative AI in consumer media; (2) A co-speculation activity where participants designed labels that would inform future consumers on media that have used Creative AI in their production. The first phase of the workshop which focused on feelings and perception featured three key takeaways: (1) consumers are willing to buy AI created works of art if the price and quality match their perceptions of how much effort went into the work created, for example, a printed painting mimicing watercolor will be percieved at lesser cost than a large canvas oil painting (ref. 4.2.1 and 4.2.2); (2) consumers are heavily biased against AI, for example, they will pay less for a painting created by an AI compared to a similar work by a human as they prefer the former to be of lower effort (P_5: *"If the content is good, I'll still accept it. But I wouldn't pay as much as I would for something [made by AI]."*), or perciving AI produced works as inferior (DP_2: "I would pay less than the other parts of that series." on book written by a human vs an AI), though the origin of this bias is unknown (ref. table 3); and (3) consumers are still trying to figure out where they stand on this situation and are grappling with the ethics, morality, and power of AI, such as for example not knowing if they would accept AI replacing writers of journalistic publications, or more broadly if AI creating media would align with their morals and what those morals even are (ref. 4.2.4). The second phase which focused on speculative label design also provided 3 key takeaways: (1) Consumers want to see AI usage Information for the media they consume (ref. 4.3.2), (2) participants preferred labels that contained more information as opposed to simpler/minimal labels, (3) Consumers are influenced by existing labels

[11] Participants are referred to as 'P_X' from hereon where X refers to the participant's number which can be found from table 2.

that they have seen causing them to expect similar designs for the AI labels.

Themes	Codes	Coding Frequency
What do consumers value?	Quality "Content" [1]	18
	Money	6
Consumer Perception of Art	AI created art is inferior or not art at all	16
	There's a human element to art	6
	Will purchase AI-created media	12
Attitude towards AI	Biased towards humans	3
	Biased against humans	3
	Biased against AI	12
	Does not trust AI	7
	Does not care if AI creates art	11
	Biased towards AI	6
Ethics and Morality	Grappling with the ethics and morality of AI created art	15
	Uncomfortable or concerned with the power and influence of AI	14

[1] Here content refers to the contents of any kind of media being consumed.

Table 3: Themes and codes from thematic analysis on Workshop discussions

4.2.1 Consumers Claim To Not Care But Show Bias Towards AI. All participants expressed, in various manners, the same sentiment, i.e., they do not care if what they are consuming was created by an AI or a Human, as long as it has good *content* and/or is worth the price being paid for it (this has been discussed in details in section 4.2.2). However, during the discussions, it was also observed that participants are in various stages of assessment over the implications of using AI to create art and how they would react to it. P_4 stated that they would not make a decision on this subject without thinking about it, and after a week of deliberation was OK with consuming works that were created by AI, stating further that unless they were looking for works by a specific person, they would not care if it was created by an AI. P_1 and P_2 shared that they would buy various works created by AI, though P_1 suggested that the contributions (in this specific case of a deceased author) should go to the author's estate (a sentiment agreed by P_7 though they had a stronger response), while P_2 stated: "I seem to not care about the legal aspect because I do not know how to attribute creative stuff....I do not understand the model...". Stronger responses in a similar sentiment included those from P_7: "I probably wouldn't care too much...", P_3 : "If I like something I don't care whether it's done by artist or it's done by AI." However, despite this *"don't care"* attitude, participants were very much grappling with the reality of the situation that AI is disrupting activities that could once solely be attributed as an innate human quality. P_5 stated: "I would buy it. If it's not a real person.", a sentiment that was echoed in a significant amount of responses, stating that the fact that AI could just create human faces and likenesses out of thin air did not sit right with them, although they responded with "... not creepy..." when asked if such behavior creeped them out, which indicates that the participant is still thinking about this aspect but cannot nail down what they find unsettling. The same participant, though certain that they would not have a problem buying a painting during DP_1, said "Now I have a problem!" the moment the floor to discuss DP_2 was opened. P_1 was still trying to process and thinking for an answer out loud stating: "... I haven't really ... wrapped my mind.... Would I OK with it or not?" while discussing DP_3, thinking specifically from the angle of their profession (journalist) and something personal to them (*viz.* a hypothetical reporting of the war in Ukraine where AI was used to

write the articles). These responses indicate that participants are still grappling with the ethical and moral aspects of the use of Creative AI, possibly with a large portion of their dilemmas sprouting from their sentiments, biases, beliefs, and the like. The results from this section show that what consumers say and what they do (including in the real world with real stakes) are different and might be contradictory and circumstantial. Hence these behaviors and attitudes must be studied in detail in the real world if there is an aim to understand customer (or more broadly societal) behavior and attitudes and how it might affect the markets, industries, and businesses that use or implement Creative AI.

4.2.2 Quality Of The Content And Money Together Power Consumer Decision Making. All participants across various discussion points indicated that they would purchase/pay for media works if the quality of the content or its monetary worth makes sense to them. When participants discussed the monetary aspects, their responses came from a personal sense of their attitudes towards purchase behavior in general. For example, P_1 and P_4 talked about how they could relate to the *'broke student'* trope, with P_1 stating that they would buy something that is "really nice" and "affordable", and P_1 saying "... I will probably buy it if I like it." during DP_1. Similarly, P_2 said "...since it says it's considerably low. I would consider buying it." P_3 said they would buy a painting if it was "very cheap and affordable" during the discussion prompt. P_7 stated "If it's good looking and I feel like it's going to really look good in my bedroom, I have to put it. I am not going to care if it's generated by AI or if it's from a very famous artist.", seemingly the only statement to equate the quality of AI art and human-made art. Participants also suggested that if using AI could bring down the prices of the works, such as lowering the price of a painting or a lower subscription cost, participants would very much be in favor for the use of AI in creating media works. For example, while discussing DP_4, P_5 stated that if the show was just mediocre it would make no difference to him if AI was used or even if it replaced some human creatives and further saw it as a cost-saving measure that the studio could implement to continue production, as opposed to if the show was an "acclaimed series before it [AI] was introduced". A further bias towards the content can be seen by the fact that all participants wanted to purchase the final AI-assisted book in DP_2, with P_7 saying "Since you mentioned Harry Potter, I'm still gonna go for it" saying that if they are a fan and would have to wait for the next Harry Potter book then they wouldn't even think about it, a statement that is consistent with what they said about purchasing an AI created painting mentioned previously. Similarly, P_6 said "If you are truly a fan of some series you never even [think] whether it's been written by someone or some robot or something." It is interesting to note that though the medium changed from paintings to books to YouTube videos, this behavior towards money and content quality did not change. For example, while discussing DP_5, P_6 said they would continue watching AI-produced videos and that it "doesn't matter." Similarly, P_7 stated that as long as it was still the same quality and worth their time, they would continue watching those videos, the purpose of those videos was "entertainment" and if that purpose was fulfilled "I'm not gonna dig into too many details". P_2 also agreed with the consensus and would continue watching the videos, going further and adding that the use of AI is still ultimately based on the creator's creativity. However, this content bias also includes a bias towards AI. For example, P_3 said "I think I want to know what's going to happen.... if there is some climax going to be there I don't mind AI writing it...", however following it up with "If I read the book first and like it then I will leave it ... But if it's not nice then I'll be blaming that book", agreeing with P_5 who stated: "If it's not good then you're like, yeah, it's all because of AI."

4.2.3 The Human Touch. Throughout the discussions, it was apparent that participants valued AI-created works as inferior to human-created works. This was very apparent during the discussions of DP_1 and DP_2. P_4 said "... randomly looking for something to decorate, yeah you can buy it." referring to the fact that unless you are looking for something special, the creator of the art does not make a difference. P_5 contrasted humans and AI creating artworks by stating that in the case of AI "... it's not a skill." because the AI is "basing it on already people who have done it." The inferiority factor continues into purchase behavior where participants remarked that they would pay less for work that

has been created by AI. P_5 stated: "If the content is good, I'll still accept it. But I wouldn't pay as much as I would for something [made by AI]." during DP_2. Similarly, P_7 remarked that they agreed that there are some costs and human labor involved in generating and perhaps making prints but wouldn't pay too much money for something they could do at home, in reference to AI-generated artworks during DP_1. During the same discussion, P_2 remarked that they would not pay a lot [for the AI paintings] and would consider buying "since it's considerably low". P_2 also held onto this purchase behavior during DP_2 stating "I would pay less than the other parts of that series." when referring to purchasing the final book written by an AI, even though it was 70% and not the whole book. P_4 also stated that "I prefer not to know it before reading the book..." since they felt that they would judge it more negatively if the content was written by an AI and did not meet their expectations. This discussion also gives rise to the understanding that consumers perceive AI-created works as inferior to human-created art. This kind of inferiority of AI art (or superiority of human creations) could be attributed to the handmade effect [33] and consumers could perceive human-made creations of higher quality, taking more time, and would be willing to pay more. However, another factor that attributes to this kind of perception could also be due to the "human" creators themselves. During the discussion of DP_2, P_5 stated that the series was based on the reputation of the author and "... what the author initially did..." and if an AI were to take over the work of the author "... it doesn't give that entire feel [and] connection to the artist." P_1 stated "... if I was a millionaire and the price wasn't an issue, pretty sure I would go for fine art ... and that would be a painting by an artist with a history." P_7 said of AI completed works "... there no human aspect in that, so they [the fans] will have that emotion and sentiment." when talking about Schubert's AI completed symphony [72].

4.2.4 Discomfort From AI's Power And Capabilities. From the discussions, it was also evident that participants were uncomfortable with the power and capabilities of AI. Throughout the workshop, P_5 was constantly emphasizing how they were put off by (and did not want) AI creating human figures, further stating multiple times that they wanted to be informed by "... some disclaimer saying this is created by AI", a request shared by P_2 in a different session stating "... there should be some kind of attribution..." when asked if they considered use of AI to produce art as stealing or inspiration. P_5 also stated, while discussing DP_4, that they were not comfortable with AI-created biases which could "... sway populations with AI, if it's not regulated..." and that if articles created were not neutral "It can go way out of control."; following which P_4 agreed saying "... as long as it is informative, then I can trust AI." recalling using ChatGPT had only been trained with information up to a certain year and some information was thus misleading. P_4 also stated that it would be nice if "They clearly show the database that the AI has been trained on so in that way you can kind of get closer to this. Maybe it's the database itself that is biased?" P_5 also compared the pre and post-eras of Elon Musk's Twitter acquisition [22] to the behavior of AI in the context of censoring certain views based on what a group of people determines, and said "But if you completely take away that option of agreeing or disagreeing from the people, then you're getting into the territory of imposing structures where you cannot see information beyond this point.", suggesting that though humans are all biased, if AI enforces such biases, it would obfuscate opposing information and provide preference to select narratives over others. P_1 while not directly stating their discomfort had stated "Right now I would not like to read an article written by an AI..." for sensitive subjects such as the war in Ukraine.

4.3 Co-Speculative Label Design

The second stage of the workshop saw participants create speculative future labels to distinguish AI-generated content. Scanned worksheets can be found in Appendix D.2. Below are the findings after performing content analysis on the combined data of the worksheet and participant statements during the discussions of co-speculation.

4.3.1 Target Groups For AI Labels. All participants stated their labels were targeted at consumers, while some participants included additional target audiences such as artists (P_3, P_5 and P_6), stores (P_6), industries (P_3), or anyone

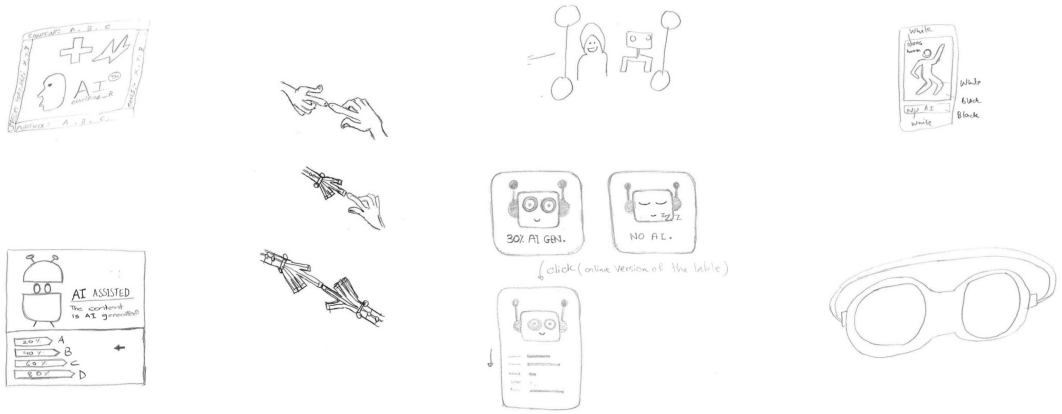


Figure 3: figure
Snippets of Pictograms from the labels designed by participants.

providing media content or services (P_5). 4 of 7 participants stated their labels intended to provide information to consumers on AI use; an additional P_5 stated that their label was intended for "software creation attribution". P_1 did not provide any data, and data provided by P_3 was extremely vague and thus unusable. It is interesting to see that, although artists and software developers are consumers, these *'creators'* are those whose professions are directly being affected by Creative AI. Perhaps this could imply that consumers might be empathetic towards them unconsciously (in opposition to what was stated by both the survey respondents and the workshop participants). Targeting labels at entities other than consumers or creators, though an interesting take by participants, does not fit the scope of this study; However, it is interesting for the broader research in label design to study how labels might be targeted towards, for example, industries and stores, and what implications they might bring.

4.3.2 Analysis of Key Design Elements in Speculative Labels. Analyzing the labels designed by participants we can understand the three parts of the labels that participants thought were important: Pictograms, 'AI Usage Amount', and Colors. Additionally, the 'layouts' of these labels were also analyzed, though they were not discussed by the participants themselves. These four particulars are described below in the order that they were deemed important from participant drawings and statements.

Usage of Pictograms. All 7 participants used some kind(s) graphical elements (such as pictograms and icons) in their label designs (ref. fig. 3). Five used some kind of robot drawing to symbolize AI, while P_5 used a spanner and gear to symbolize AI. On the other hand, only 3 of the 6 labels used some pictographic representation for *human being*. It could be understood then that consumers would more readily understand a label denoting Creative AI use if it contained a pictogram of a robot or something similar to denote 'AI Use'. Graphical elements were also the largest (by area) and most visually directed part of the entire label. P_2 used the *node* similar to those used to describe neural networks. The design from P_3 was an outlier where the entire label was a singular pictogram, further indicating that pictograms were considered to be the most important aspect of a label's design to the participants.

AI Usage Amount Information (AIuAI). 6 out of 7 participants indicated *AI Usage Amount* on their label designs. This term includes both numerical values (such as percentage of product created by AI), and *blanket* terms such as "AI-assisted", "Fully AI Produced", etc. Participants chose different manners in which to convey these terms; two

participants used only graphical elements to convey this information, while the remaining 4 participants used both graphical elements and texts (with P_5 using color as well). All 6 participants also indicated that their *scale of AI usage* would range from "No AI Used" (or similar text) to "100% AI Produced" or (similar text). Except for P_5, no participants designed labels that had "Human Created" or similar text, though multiple participants included labels that contained *human* figures. 3 out of the 6 participants included *percentage* rating and letter grade ratings to signify how much AI was used, with one other participant using only percentage and no letter grade. The remaining participants who did not include any text chose three forms of labels: Fully AI produced, AI Assisted, and No AI Used. While they chose the same 3 forms of labeling, they used different words to indicate what their labels meant. P_1 used the phrases "Entirely Man Made", "AI + Man Made", and "Entirely Made By AI"; P_2 used the phrases "No conscious help from LLM", "AI Assisted", and "Fully AI Created".

Color. Though color was not provided in the reference labels, some participants attempted to include color in their designs. P_5 and P_7 used gradation-based color[12] labels to depict the amount of AI used in their label designs, similar to the use of color in the EU energy label[13]. P_5 used colors to provide a visual indication on how much of the *product* used AI. The design utilized a green-to-red gradient, where (generally) green is *highest good* and red is *highest bad* [71], though interestingly P_5 used green to represent 100% AI use, possibly indicating that they saw AI use as a net positive.

Label Layout And Structure. As stated before, the graphical element was the centerpiece of all 7 label designs taking up the most space. 4 of 7 participants placed their label elements inside rectangular boxes. 2 of the 4 had placed the label at the top of their designs, while the other 2 had placed them in the center. AIuAI headings (such as percentage and/or grades) were the second largest aspects of the labels, though their placements about the graphical elements varied greatly. There were also two particularly interesting layouts: P_6 included information on the types of AI tools/technologies used and audience information in the border, and P_4 included a version of the label that could be used in digital environments where the label could display more information when expanded. Hence future label designs should include large and easy-to-understand pictograms with AIuAI information as the most prominent parts of their label designs, while also looking into the areas of application.

4.4 Changes In Perception

Prior to the start of the workshop, participants were made to fill out a nine-item questionnaire that corresponded to their attitudes and perceptions on topics surrounding Creative AI. An identical questionnaire was provided to the participants at the end of the workshop to evaluate if their attitudes and perceptions changed after learning new information. Overall, there is an average improvement of 0.06 points (or 1.95%) positively towards Creative AI. Looking at the individual per-question scores, however, we can see that a majority of attitudes remain unchanged. Table 4 below shows how the averages for each question changed between the pre and post workshop sessions. Tables 8 and 12 in appendix D.3 contain the question codes and full overview of the answers changed at a participant level.

[12]The drawings are still in black and white as no color drawing materials were provided to participants.
[13]A reference image of the EU energy label can be found here.

Question	Pre Workshop									Post Workshop								
	A	B	C	D	E	F	G	H	I	A	B	C	D	E	F	G	H	I
Average	2.57	3.57	2.57	2.71	3.86	2.57	3.29	3.43	3.14	3.00	3.57	2.57	2.57	3.86	2.43	3.29	3.43	3.57

Table 4: Comparison table of question averages pre and post workshop

Question A asked if consumer choice labels affected their purchase decisions in the past and increased by 0.43 points (16.73%). There doesn't appear to be a concrete reason for this, but perhaps might be due to participants realizing that labels did unconsciously affect their decisions. **Question B** evaluated if consumers would like to be informed of Creative AI use for media they pay for and remained unchanged with a score of 3.57 indicating moderately high importance. **Question C** evaluated consumer spending habits around content that is fully produced by Creative AI and remained unchanged with a score of 2.57 indicating that on average consumers would pay less for content produced in such a way. **Question D** evaluated consumer spending habits around content that is partially produced/assisted by Creative AI reduced by 0.14 points (or 5.17%) indicating that consumers would now more decisively pay less for 'AI-assisted' media. **Question E** evaluated if consumers thought they would see Creative AI used more in the media they consume, remained unchanged with a score of 3.86 indicating that a majority of consumers are cautiously positive about this trend. **Question F** evaluated if consumers saw Creative AI as a threat to human creativity, and decreased by 0.14 points (or 5.44%) indicating that fewer participants saw AI as a threat to human creativity with only somewhat average agreement. **Question G** evaluated if consumers saw Creative AI as a threat to creative professions/professionals, and also remained unchanged with an average score of 3.29 indicating that consumers are mostly unsure. **Question H** evaluated if consumers saw some creative jobs being at least majorly replaced by Creative AI, and remained unchanged with a score of 3.43 indicating that most consumers believed (to a fair degree) this to be true. **Question I** asked participants if they would be okay with some creative jobs being replaced by Creative AI, and increased by 0.43 points (or 13.69%) indicating that more participants agreed that they would be fine with more creative jobs going away.
Since the reasons for these questions were not directly requested, it is unknown why the specific responses to questions changed. However, it can be understood from the workshop discussions that overall participants are (1) unsure of where they stand concerning the ethical and moral quandaries of Creative AI, and (2) are open to changing their opinions when they learn more.

4.5 Recommendations For Consumer Labeling Based On Workshop Insights

From the initial discussions phase of the workshop, it can be seen that consumers are willing to spend money on AI-created media, though there exists a dichotomy of bias against AI and AI-created works; another way to phrase this dichotomy would be 'indecision'. Consumers are indecisive and still attempting to understand where they stand on the whole issue of AI (Creative or otherwise) and the effect it has on people (ref. 4.2.1). How consumers react in the long term once media using Creative AI becomes far more mainstream than it is right now to be seen, though from the results of the workshop it is apparent that providing consumers with information related to AI usage can enable them to make better and more informed decisions. The label designs have indicated that for the most part, consumers would like to see AI Usage Information. The reason for this ranges from simple reasons such as curiosity or consumer choice empowerment to more sensitive reasons such as personal connection to the media/content/material (such as AI writing news articles about war), fear of AI's power and misinformation, and preventing AI from influencing their biases and

world views. The workshop suggests that this information be presented as either quantitative (percentage or grades) or qualitative (Human Produced, AI Produced, etc.) measures. These measures can be depicted or aided by pictographic elements, primarily that of robots or other such pictograms that depict AI, and perhaps also using figures that depict people. These pictograms (or similar graphical elements) are the centerpieces of the label designs and the most visually prominent part of the designs. Apart from this, color might also be an important aspect of future label designs, though the workshops only yielded preliminary evidence of this. Comparing the labels with each other, we can see by the voting dots that participants preferred more informative labels, *viz.* labels that provided more information about AI usage. Participants also seemed to prefer labels that had well-represented pictograms where it was obvious to know who the creator was (AI, Human, or collaboration) by looking at the label. The label designs designed by participants also bear some resemblances or are inspired by certain pre-existing labels (such as the EU energy label discussed previously), which means that more familiar label designs can be studied not just to design the labels themselves but, for example, localize the labels to different markets, medium (physical or digital), or media type. Though these results have provided vital information on the recommended routes to take while designing such AI labels for consumers, these results must be studied or worked upon further due to certain drawbacks and limitations discussed in the next section.

4.6 Meta Analysis of N-Speculation Workshop

To provide a complete picture of the results above it is important to also address the possible drawbacks and biases that might have affected the study results. First, all the workshop participants were pursuing higher education at a technical university and would thus have more insights about Creative AI than the average consumer, possibly resulting in volunteer bias. However, the fact that the results from the survey, which saw respondents from all over the world and from various educational backgrounds, still concurred with some of the workshop results provides validity to the workshop results. When it came to label designs, pre-existing labels might have caused participants to unconsciously draw inspiration from or mimic existing designs. Further, since some examples of AI labels (ref. fig. 9 in Appendix D.4) were shown, this could have caused participants to narrow their thinking and be influenced on what these label designs should look like. Further, the 15-minute time limit on designing the labels might have caused participants to depend on the existing label designs shown. Additionally, these pre-conceived examples in combination with the fact that participants *had* to design labels could have caused framing bias where labeling might have been presented in a favorable light; while the survey does provide major confirmation of the validity of this data, the option to dismiss labels as anything necessary was simply not provided during the workshop and, as can be seen by the survey, was an option that was chosen by not an insignificant number of respondents. However, seeing the usage of color and variations of other kinds of graphical elements in label designs that highly differ from those presented means that though participants might have been influenced by the designs presented, they were not heavily biased to creating designs that copied those presented. Apart from bias, it can also be seen that the scope of these label designs was limited due to the constraints of time and limited variables being studied during the study. For example, two participants indicated their intentions to use color in their label designs; it is thus possible that this number could have been higher given (1) more time, and (2) coloring tools (colored pencils, crayons, etc.) to implement color in their designs. This could affect how label designs using or not using color could be perceived and how the factors discussed above could have their order of importance rearranged. The scope of these labels is also limited to the medium (paper) that was used during the workshop; perhaps holographic labels, labels with texture, or even video labels might have greater or lesser effects on consumers in physical or digital spaces. There are plenty of possible factors that could enhance these AI labels, however, it is in the nature of studies to be delimited and provide directions for future research work to be carried out (*vide* 5.5).

5 DISCUSSION

The results of this study provide preliminary insights into the perceptions and attitudes of consumers surrounding the use of Creative AI in media, and positively studied a viable solution to the problems related to consumer choice that will arise. Through the content analysis of 32 Creative AI tools, it was discovered that Creative AI, though capable, is not a replacement for human professionals anytime soon, though some jobs/tasks will be replaced. The survey helped assess the various attitudes and perceptions of consumers on various aspects relating to the use of Creative AI in media. The survey showed that less than half of consumers wanted to see any kind of compensation for the loss of livelihoods due to Creative AI though interestingly 45% of respondents either did not know enough to be definitive or did not want to admit what they really thought. The survey also showed that consumers care about the state of the information they receive and are aware of the impact that Creative AI could have on factual accuracy and reporting. Finally, the survey showed positive preliminary evidence of the need for labeling AI use in consumer media. The N-Speculative Workshop expanded on the results of the survey and provided insights into the design for future AI labels. The N-Speculative workshop showed that though consumers claim to not care about AI-created works they also have underlying biases towards AI. Further, it also showed that consumers do not completely trust AI, especially with content that is personal to them or content that requires high factual accuracy. Similar to what was seen in the survey, the N-Speculative workshop also showed that consumers are aware of and concerned with the power that AI possesses when used in media. The N-Speculative workshop also brought up new revelations; consumers would highly take into consideration the price, quality, and creator (Human or AI) to evaluate if and how much they would pay for certain work, valuing AI work as inferior to human work. The N-Speculative workshop also provided preliminary insights into the design of AI labels. Prominent pictograms and detailed AI usage information topped the list of requirements for consumers. It was also interesting to see that consumers are influenced by other pre-existing labels (ex. P_5 and P_7 drew inspiration from the EU energy label) and future label designs should take those into account. Finally, the N-Speculative workshop also showed that there is a need to empower consumers with more information for them to make more informed choices in the era of Creative AI.

5.1 Advancements In Creative AI Is Dependent On Educating The Consumer

Looking at sections 4.2.4 and the responses in the Survey Results (4.1) we can see that people have a variety of different understandings of the way AI works or will affect them. Such confusion also creates hindrances to the implementation and acceptance of AI in mainstream creative uses. Further, once AI-created works become mainstream, they may reorganize the consumer purchase market, where AI-created works might take over the *'fast-markets'*, while human-produced works would take up the higher-end or luxury markets; analogous to the consumer perception of mass-produced fast fashion such as H&M or Zara and luxury brands such as Armani or Gucci that pride themselves on their *handmade* craftmanship [66, 90]. We must venture into a reality where AI-created media becomes a significant role in our lives along with consumers. It is thus important for a variety of reasons that there be a dissemination of information related to Creative AI in the hopes of alleviating some of the concerns that scare consumers.

5.2 The Future Of AI Labeling

Prior to commencing the study, it was unknown if consumers would be positive towards AI labels. However, given the undecided or mid-point responses from both the survey respondents as well as the "still thinking" nature of the workshop participants, we can see that labels might help give them more information about their purchases or where and how they spend their money. It is currently not possible to predict if in the future there will be a requirement for AI labeling without any large-scale and long-term studies into consumer behavior surrounding Creative AI and its uses.

However, these studies will also have to wait until Creative AI becomes more mainstream in media. Thus, this thesis, presents a preliminary window into a mansion of studies that is Creative AI and consumerism, and more specifically into AI labels. The current results of this study indicate that consumers are still unsure about Creative AI (mostly) due to the information gap that exists between consumer perception and reality, which leads to bias, uncertainty, and fear. However, the study results also indicate that with *education* on Creative AI, a change in such perception and attitudes including purchase behaviors is positively witnessed. Further, the results of the N-Speculative workshop present AI labels as a solution in a favorable light and provide indications of the potential characteristics that these future AI labels should have to be effective.

5.3 Ethical Discussions Surrounding Of Creative AI And Consumer Choice

There has been quite a lot of discussion surrounding the ethics of using Creative AI. These questions have not only been reserved for philosophers or academics but also the laypeople (consumers) who, though lacking any specializations in these areas, are pondering these moral and ethical quandaries. While nobody has the definite answers to these questions yet, it is important to have discussions of the ethics that surround Creative AI to ensure responsible implementation and research, free from hasty decisions such as calling for a pause of AI development [63], replacing human works with AI tools [84], or outright predicting the downfall of companies that embrace AI [75]. Hence this section surrounding the ethics of the subject matter is centered around the following premise: "What are the ethicalities of allowing consumers to steer the course of Creative AI technology, whether it be mainstream adoption, regulation, or its demise?"

5.3.1 AI Labels As A Counter To AI Bias. Bias in AI systems has been well documented and reported for a long time; Google Photos labeled a black couple as gorillas [55], Meta's Galactica started writing racist and inaccurate scientific literature [28], and an image generator providing racist and sexist images [4] are just some of the recently reported incidents of AI producing biased results. This concern of bias was also shared by the workshop participants (ref. sec. 4.2) and the survey respondents (ref. sec. 4.1). AI labels might serve as a way to at least restrict the amount of damage caused. For example, labeling sensitive content, such as coverage of war or ongoing societal protest, that has been written by Creative AI and providing the warning of possible bias or information of the relevant *weights* used by the model could be a solution of ensuring that consumers are well informed about any potential biases that may have crept in, rather than being (or believing that they are) misled and blame Creative AI.

5.3.2 Consumer Perception Of Artistic Identity. Both the survey responses and the workshop discussions saw consumers state that more than the work itself, the name of the artist/author held meaning to them. Before the explosion of Creative AI, artistic creations were always tied to a human creator and the styles reflected in these works became *trademarks*, or a way to recognize the artist from the art: This could be, for example, Wes Anderson's use of symmetry, Salvador Dalí's surrealist painting style, Pablo Picasso's Cubism, Andy Warhol's Pop Art prints, or ad-libs found in rap music. Since Creative AI generates works based on an amalgamation of its learnings, it does not have a distinct style of its own. Thus, the question arises: Is it still art when the "artist" element of a work is stripped away? Does the art now become superficial? People purchase paintings, go to concerts, or buy a creator's merchandise because of the name(s) that are attached to it; would consumers buy *"A painting by Mid Journey"*, or watch a movie *"Written and Directed by RunwayML"*? Further, if consumers no longer care about the identity of the artists, would it spell the same fate for the artists as it did for carpenters and artisans who had their artistic identities and livelihoods diminished by the first industrial revolution? How many names of famous artisans and carpenters do we know today as compared to famous musicians and painters? These questions become more difficult when applied to forms of art (such as acting) where the art and the artists *'cannot be separated'*; How much of their likeness does a person own, is it only their face,

their whole body, their voice [81], or some combination of these? Do people lose *copyrights* to their likeness after their death similar to other copyrighted works [83]?

5.3.3 Consumers And Deciding Ethics Around Use Of Creative AI. The results of this study clearly show that consumers are still attempting to understand where their morals lie on the use of Creative AI, in media or otherwise. However, is it ethical for consumers to be deciding the ethicality of Creative AI? What would happen if consumers decided that all Creative AI is bad? Should technological evolution be halted simply because a large number of people deem it to be unethical in manners that resemble those of the first industrial revolution? Is it ethical for businesses and organizations to force consumers to accept Creative AI-produced work as the new status quo? Consumerism is a highly social phenomenon and consumer perceptions might be affected by societal acceptances or common social attitudes. For example, wide-scale adoption of smart glasses was impeded due to social factors such as rudely referring to the wearers as "glassholes" [70]. The reverse can also be seen with, for example, the wide adoption of ChatGPT among students [82]. Consumers' perceptions can be significantly influenced by information that is readily available and easily understood, such as social media, mainstream journalistic media, or social discussions. Hence it is not *out of left field* to suggest that these perceptions or understandings might be flawed or biased in comparison to scientific literature, though given that these sources are far more accessible to the everyday consumer, they are often the go-to sources of information. This statement then loops back into the core ethical question of this paper: "What are the ethicalities of allowing consumers to steer the course of Creative AI technology, whether it be mainstream adoption, regulation, or its demise?" It must be made a priority for all stakeholders, academics, creators, businesses, and consumers, to approach the subject of Creative AI with an open mind and educate themselves and others with unbiased and high-quality information that, in an ideal world, the era of Creative AI shaped by clear thinking individuals and organizations without crude and misguided regulations, rather than agenda driven corporations and interest groups or misinformed consumers.

5.4 Reflections On Sustainability In The Creative AI Era

Creative AI is set to change the landscape of not just Creative work, but also our relationships with business, our impact on the environment, livelihoods, inequality, education landscapes, and landmark legal decisions surrounding our usage of technology, art, and copyright...to name a few. Most obvious among these, is that Creative AI provides a lot of (mostly monetary) incentives: Compared to human workers, AI costs less [56], requires far shorter *downtime* as AI does not need to sleep or rest, and is (on average) faster than humans; AI is also not governed by the same laws that human workers are, such as maximum working hours or mandatory leaves, meaning that companies can use their AI systems as they please without the *'hassles'* that come with human workers. Such advancements are very likely (*cf.* sec. 1) and would start with mass unemployment in some professions, snowballing into greater implications on social sustainability. Social sustainability can be defined as managing the positive and negative impacts of systems, processes, organizations, and activities on people and society [9]. Extrapolating from the above situation of mass Creative AI adoption, we can see that mass unemployment will eventually lead to some professions being "AItomated". This will lead to mass protests, similar to those seen during the 2023 Hollywood writers [10] and actors [76] strikes, though the outcome might not be similarly favorable. Such volatility of some careers will result in mass shifts to more in-demand professions such as AI or ML, resulting in some academic streams seeing greater competition and importance, while others such as creative arts or design will see a decline. Such shifts might also result in widening inequalities, as fields of AI/ML are often white, and male dominated [17]. Such changing paradigms will affect people en-mass and might see regulations being made to compensate for these 'human losses', which might stifle advancements of Creative AI especially if business bottom-lines are hit. Mass adoption of AI does not only affect social sustainability but also has

significant environmental impacts. One estimate suggests that simply the creation of GPT-3 is estimated to have emitted as much carbon as a hundred gas cars in a year [68, 79]; AI, from training to active deployment and use, is very energy intensive and will require the setting up of new green energy infrastructure (which will against have financial and social impacts) to meet the energy demands sustainably.

5.5 Limitations And Future Work

Creative AI is a very broad field. Not only is the scope of research extremely wide, but the lack of a concrete definition also means that there is going to be a large grey area for tools and technologies that sit on the fence of being classified as Creative AI. For example, AI used to perform content-aware fill in Adobe Photoshop or After Effects could be considered to be Creative AI, if the replacement and removal procedures are considered to be *Creative*. Further, the new AI tools are rapidly being created and the capabilities of the old ones are also being expanded. For example, during the time of this study, ChatGPT was expanded by using GPT-4 and even got plugin support. It is of course without doubt that the expansion of this definition and taking another look at these Creative AI tools will lead to changes in this study's results, for example, by including Creative AI tools that do not have easily accessible versions. The N-Speculation method, despite its advantages over traditional design speculation, also possesses certain drawbacks. At its core, N-Speculation is still a type of speculative design approach and as a result, comes with some uncertainty Further, N-Speculation aims to act as a bridge between traditional design research and speculative design research, the consequence of this placement means that the results obtained are not as conclusive, however provides preliminary (or even actionable) insights to inform future studies around the subject matter (unlike traditional speculative design). This same lack of conclusiveness is also applied to workshops, where it can safely be predicted that there will be differences in what people say (or do) during the workshop and what they end up doing in real life, such as for example, when their own money is at stake. Further, the workshop consisted of only 7 participants who all have similar insights on the subject matter based on their educational qualifications and future studies must expand on the number of participants as well as the diversity of the study group.

6 CONCLUSION

This study has provided initial scientific insights into consumer perception of Creative AI use in media and has also provided insights into the possible solution of AI labels to empower consumer choice in this new frontier. The results from the content analysis show that Creative AI tools are highly capable of replacing certain professions entirely, though the implementation of this in professional settings remains to be seen. The content analysis also tells us that current Creative AI tools are highly capable in image, text, and code areas, while other areas such as music and video continue to be developed and will only get better with time. The survey and workshop discussion results show a mostly ambivalent consumer that is unlikely to stay away from or boycott media simply because they are 'made by Creative AI'. However, the study also shows that consumers are curious and want information that allows them to make more informed choices. Further, consumers are, like any other kind of spending, going to put money and quality above most other factors. When it came to labeling the study discovered that consumers are also, for the most part, in favor of labeling media that has used Creative AI. Further, the workshop also provided insights into the important design elements of such labels: Pictograms/Graphical Elements, AI Usage Information, well-structured layouts of AIuAI and pictogram(s), and possibly color. Given the prominence and unprecedented speed of development of Creative AI and its impact on society, there a significant need to study consumer choice, bias, and misinformation, and be actively involved in some way towards the synergetic development of Creative AI for societal good.

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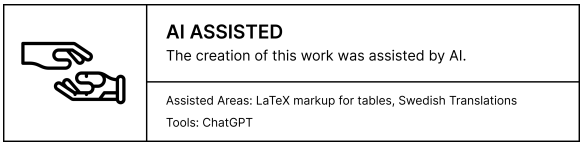
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A DECLARATION OF AI USAGE

ChatGPT was used to generate tables in LATEX code, and assist in Swedish translations.



B TABLES OF AI TOOLS AND COUNTS

S.No	Tool	Category	Transparency	Pricing Model
1	Stable Diffusion[1]	I	O	F
2	ChatGPT	C, T	Cl	Fm
3	DALL-E	I	Cl	PPU
4	Runway Gen 2	V	Cl	PPU
5	Descript Overdub	S	Cl	PPU
6	Synthesia	V[2], S	Cl	PTU
7	Jukebox	M	O	F
8	MusicLM	M	O	F
9	Mubert	S	Cl	Fm
10	AutoDraw	I	Cl	F
11	MidJourney	I	Cl	PTU
12	NVIDIA Canvas	I	Cl	F[3]
13	Jenni	T	Cl	PPU
14	AI Poem Generator	T	Cl	F
15	GitHub Copilot	C	Cl	PTU
16	Cody by Sourcegraph	C	Cl	Fm
17	Amazon CodeWhisperer	C	Cl	Fm
18	Mintlify Writer	C	Cl	Fm

19	AI2SQL	C	Cl	PTU
20	HourOne	V[2], S[4]	Cl	PTU
21	PrometheanAI	G	Cl	Fm
22	Scenario	G[5]	Cl	Fm
23	Runway Infinite Image	I	Cl	PPU
24	Adobe Firefly	I	Cl	F
25	vizcom	I	Cl	Fm
26	Maket	I	Cl	Fm
27	Poly	I	Cl	Fm
28	DeepFaceLab	V	O	F
29	Riffusion	M	O	F
30	Jasper	T, I	Cl	PTU
31	Bard	T	Cl	F
32	BingAI	T	Cl	F

Table 5: AI tools analyzed for content analysis.

Key		
Identifier	**Type**	**Count**
I	Image	11
C	Code	6
T	Text	6
S	Speech Audio	4
V	Video	4
M	Music	3
G	Game Assets	2
O	Open Source	5
Cl	Closed Source	27
F	Free To Use	11
Fm	Freemium	10
PPU	Pay Per Use	5
PTU	Pay To Use	4

[1] Various versions of Stable Diffusion exist. It was tested through Lexica art and through a local install.

[2] Generates videos with AI-generated photoreal avatars.

[3] Requires the purchase of a compatible GPU.

[4] Can also use AI audio voice cloning, though this could not be evaluated.

[5] Specialized Image creation for 2D game sprites.

Table 5: Counts and identifiers for table 5.

C SURVEY QUESTIONS, DEMOGRAPHICS AND FIGURES

C.1 Survey Questions

- Questions 1 and 2 asked respondents to rate the importance of various information related to AI use in their products/services on a scale of 1 (Not at all important) to 5 (Extremely important), for digital content and physical products respectively.
- Questions 3 and 4 asked respondents which kinds of declarative labels they would like to see, if any at all, on physical products and digital content respectively.
- Question 5 asked respondents if lost livelihoods due to AI should be compensated.
- Question 6 asked respondents which specific content would they like to see information regarding AI use.
- Question 8 asked respondents if they had heard of Generative AI before to get a better understanding if respondents understood fully what was being talked about in the survey.
- Questions 9 through 14 are demographic questions aimed at providing a better understanding of the survey results.

C.2 Survey Demographics

Response ID	Nationality	Gender	Age Group	Job / Profession	Highest Educational Qualification
105915980	India	Male	18-24	Graduate Student	Bachelors
105922236	Finland	Male	25-34	Cloud Specialist	Bachelors
105983206	India	Male	18-24	UX Designer	Bachelors
105984650	Sweden	Female	35-44	Phd candidate	Masters
105989097	Netherlands	Non-binary	25-34	Doctoral Researcher	Masters
106003131			25-34	Doctoral student in Computer Science	Masters
106167566	India	Male	25-34	Researcher	Masters
106196947	India	Female	18-24	Designer	Bachelors
106268604	Greece	Female	25-34	PhD student	Masters
107867465	United Kingdom	Female	35-44	Retail	Masters
107883789					
107937855	India	Male	18-24	UX Designer	Bachelors
107940390	Finland	Women	25-34	Student	Masters
107963000	Armenia	Female	25-34	Media Manager	Masters
107976691	Turkey			Academician	Masters
108047246	Finland	Male	25-34	Software developer	

108182889	Finland	Male	25-34	Sales	Some college
108226404	India	Male	35-44		Masters
108260980	Italy	Female	25-34	Academic researcher	Doctorate
108261938	Netherlands	Female	18-24	Student	Trade/vocational/technical
108699788					
108725433					
108847032	Austria	Female	18-24	Student	Bachelors
108867010	New Zealand	Female	45-54	Design/photography/music/business owner	Trade/vocational/technical
108873510	Algeria	Female	18-24	graphic designer	Bachelors
108961556	Nigeria	Female	18-24	Student	High school
109004560	United Kingdom	Female	18-24	Student	Bachelors
109091070	United States	Female	55-64	student	Masters
109110178					
109152700	Australia	Female	45-54	Business owner	Bachelors
109242535	United Kingdom	Female	18-24	Bar associate	Bachelors
109388034	Albania	Female	25-34		Bachelors
109388123	Croatia	Agender	25-34		Masters
109458094	Australia	Female	18-24	research assistant	Bachelors
109498070	Pakistan	Female	25-34	student	Doctorate
109504640	Japan	Female	18-24	Student	Bachelors
109726613	China	Female	25-34	music	Other
109727414	Ireland	Male	25-34	Teacher	Masters
109928125	China	Female	25-34	psychological studies	Masters
110203501	United States	Male	35-44	Engineering	Bachelors

Table 6: Survey Demographics

C.3 Likert scores for Q1 and Q2

Medium ⟶		Physical		Digital	
Codes	**Item**	**Score**	**Standard Error**	**Score**	**Standard Error**
A	To know if what I am consuming/purchasing has been produced by AI as a whole.	2.7	0.215	2.95	0.209
B	To know if AI has assisted in the creation of my content/purchase.	2.75	0.217	2.92	0.197
C	Which AI tools/technologies has been used.	2.65	0.216	2.7	0.227
D	To know where AI has been used in my content/product	2.92	0.204	3.23	0.198
E	To know if the humans replaced by AI, were compensated for their lost work.	3.00	0.238	2.83	0.226

Table 7: Likert scores of items from Q1 and Q2

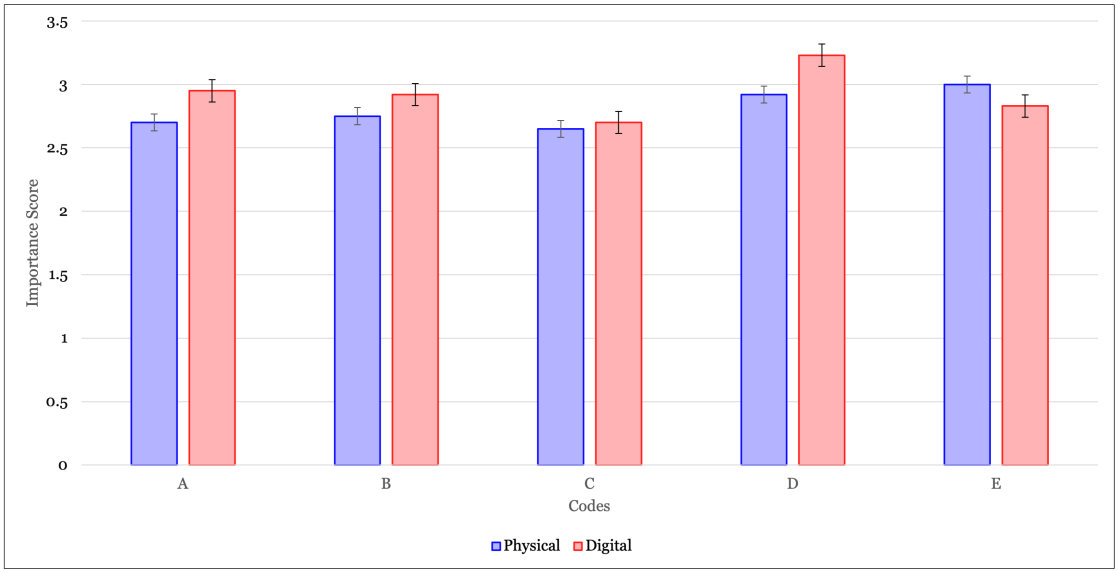


Figure 4: Visual comparison of Table 7

C.4 Prefered Declarations of survey participants

Codes	**Item**	**Physical**	**Digital**
A	AI Produced	6	11
B	AI-Free	5	6
C	100% Human Creation	9	9
D	AI Assisted	11	15
E	Does not matter, but a declaration is required	13	14
F	I don’t care about such labels	9	4
G	Something else	1	0

Table 8: Votes for the respective AI declarations made by survey participants.

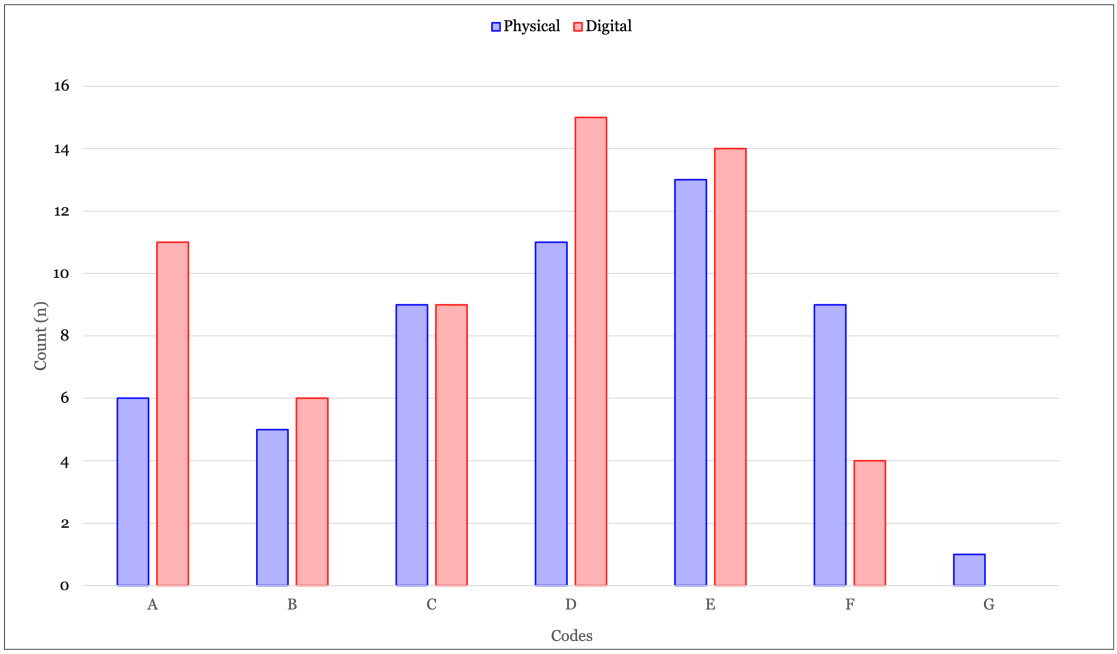


Figure 5: Chart of comparing votes for each AI declaration from table 8

C.5 Media specific AI declaration preferences

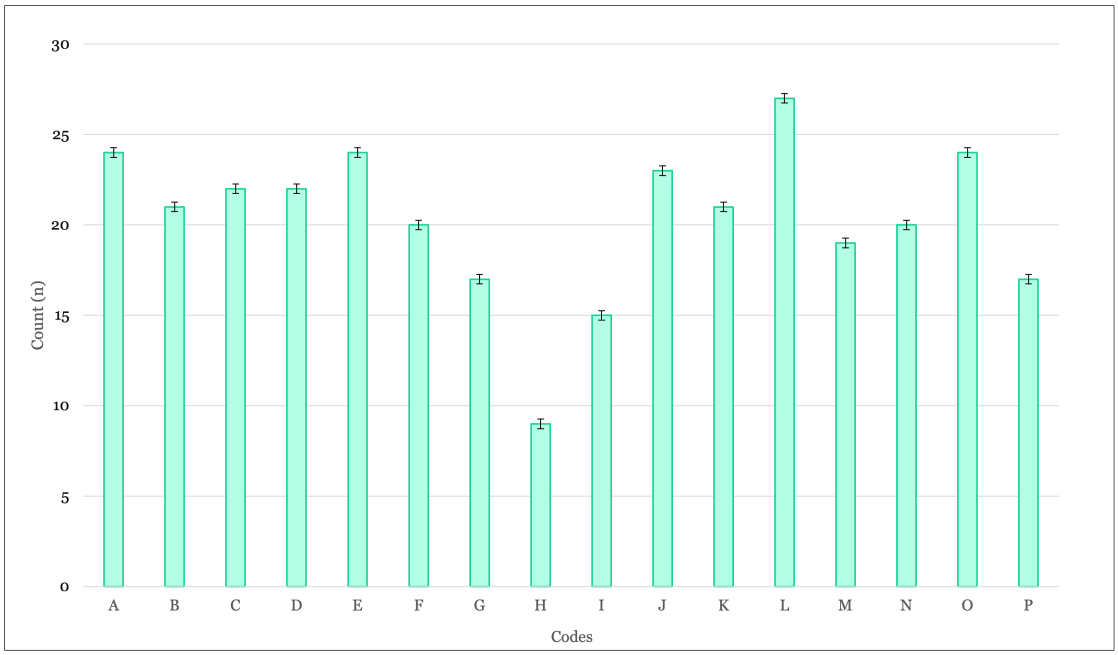


Figure 6: Sub media categories and how many survey respondents would like to be informed about generative AI use.

Codes	Item	Count
A	YouTube Videos	24
B	Music	21
C	Movies	22
D	TV Shows	22
E	eBooks	24
F	Physical Books	20
G	Comics	17
H	Video Games	9
I	Software Products	15
J	News Articles	23
K	Review Articles	21
L	Scientific Papers/Articles	27
M	Blog Posts	19
N	Celebrity/Influencer Social Media Posts	20
O	Digital Art Pieces	24
P	Physical Art Pieces	17
Standard Error		**0.263**

Table 9: Codes with count of subcategories with counts and standard error for data in Figure 6.

D WORKSHOP SUPPLEMENTRIES

D.1 Table of Prompts

ID	Prompt Title	Prompt Description
DP_1	How does cost affect consumer choice?	You want to buy decorations for your new house, so you go shopping. You come across a painting that you think will look good on your wall. You check the price tag and can't believe how affordable it is! It looks like a mistake! Paintings don't cost this much?! You check with the store employee. They inform you that the store sells paintings generated by AI and that is how they can sell them for such a low price and high quality.

ID	Prompt Title	Prompt Description
DP_2	Ethics, Fandom, and what do people look for in art?	The 7th and final book of a series that you like has been released after nearly a decade! You have been waiting so long for the cliffhanger from the 6th book to resolve, so you immediately go online and order it! After ordering you wonder why there has been such a delay, so you look it up only to find that the author had passed away after completing about 30% of the final installment (book). Turns out an AI had been trained on her older works and has completed the remaining 70% of her book.
DP_3	Artist or Information	You like to stay on top of the latest information in your area of interest (technology, fashion, gaming, entertainment, etc.). You pull up an article on an online publication (a blog or type of non-traditional news) website you frequent due to their high editorial standards. However, you notice there is something off about the article's style and language. You check to see who the author is and to your surprise instead of a name you find a label "AI Produced. Click to learn more." You click and learn that to stay in business, they have switched to using AI to write articles instead of humans.
DP_4	What if nothing changed?	You enjoy binge-watching a soap opera (formulaic TV shows that go on for years and may have daily episodes) on lazy days as a comfort show. It has a very predictable plot line, mediocre acting, and a template-esque editing and screenplay structure across episodes. During an episode you notice that the credits are much shorter than usual, a trend which continues through all the following episodes. You take a closer look to realize that no writer or editor are mentioned. You look it up to realize that the show is now fully written and edited by AI, though you have never noticed this change while watching an episode.
DP_5	Art or Artist	A YouTube Content Creator that you have been following for years has suddenly started uploading daily instead of weekly and the content quality is still as high as you would expect from them. You then find a video where the channel addresses that they will be producing some videos using AI. This includes writing, editing, and even the likeness of your YouTube creator! You can't seem to tell the difference between which is made by an AI, and which is made by the creator themselves.

Table 10: Prompts Table

D.2 Co-Speculative label creation worksheets

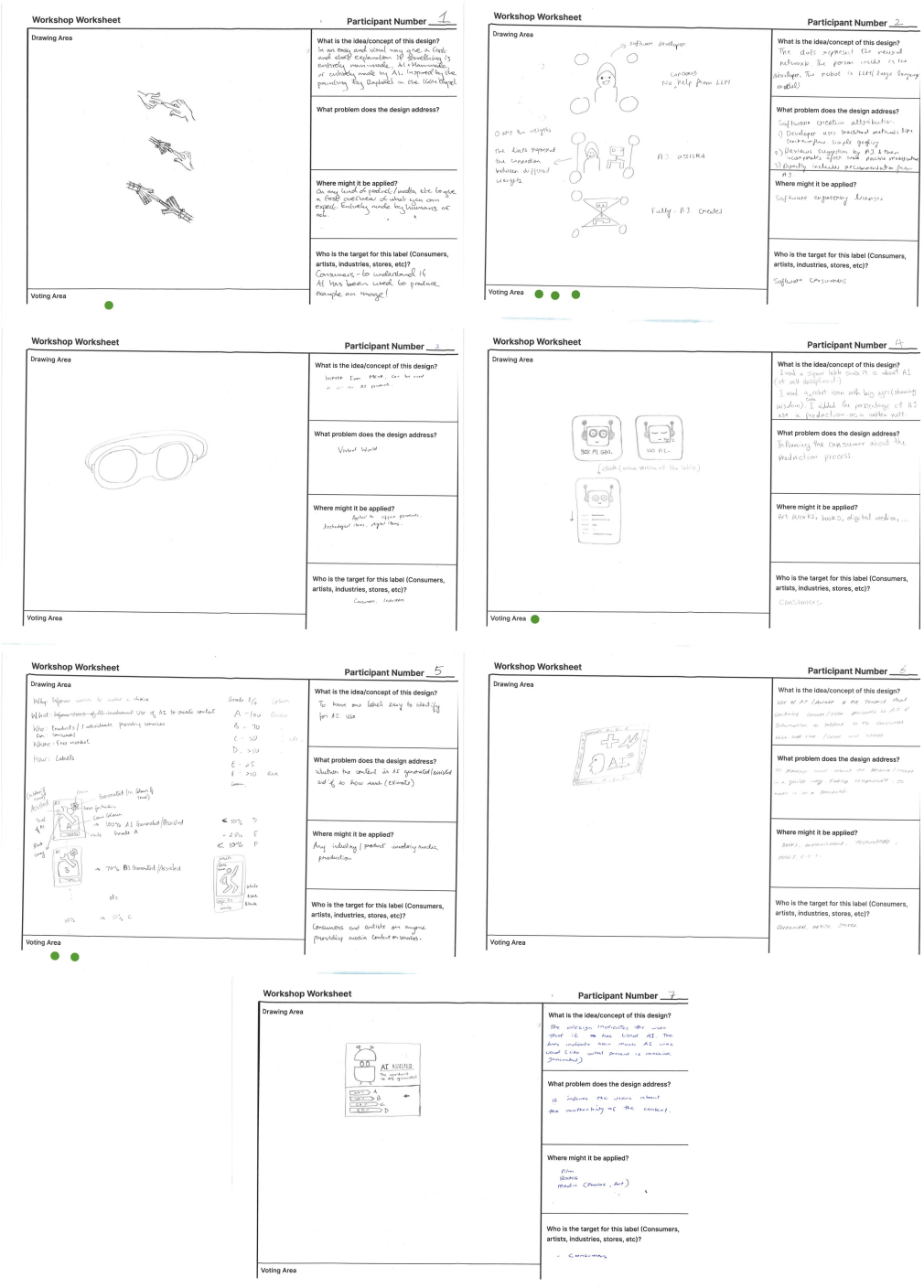


Figure 7: Co-Speculative label design worksheets Of participants.

D.3 Pre and Post Workshop Questionnaire Data

Code	Question
A	Consumer choice labels (such as FairTrade, Energy Star, 80Plus, etc.) have affected my purchase decisions in the past
B	It is important for me as a consumer, to be informed about the use of generative or Creative AI in the media I buy (or pay for)
C	When it comes to paying for media that has been completely/fully/wholly produced by AI.
D	When it comes to paying for media that has been produced partially (or been assisted) by AI
E	I will see more generative AI use in the media I can buy or consume.
F	I see generative AI as a threat to human creativity.
G	I see generative AI as a threat to the livelihoods/jobs/professions of some creative professionals
H	I can see some creative jobs being fully or majorly replaced by generative AI.
I	I am ok with some creative jobs being replaced by generative AI.

Table 11: Questionaire questions with codes.

PID	Pre Workshop										Post Workshop									
	A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**P.Avg**	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**P.Avg**
P1	3	4	3	3	4	3	4	4	4	3.56	3	4	3	3	4	3	4	4	4	3.56
P2	3	3	2	3	5	2	4	3	4	3.22	2	3	2	3	4	1	4	2	3	2.67
P3	3	4	4	3	4	3	3	4	2	3.33	3	4	4	3	3	3	4	4	3	3.44
P4	1	5	2	2	3	2	2	3	4	2.67	2	5	3	2	5	1	1	4	5	3.11
P5	2	3	2	2	4	2	2	3	3	2.56	2	3	2	2	4	2	3	2	4	2.67
P6	4	4	3	3	4	2	4	4	3	3.44	5	4	2	3	4	3	3	4	4	3.56
P7	2	2	2	3	3	4	4	3	2	2.78	4	2	2	2	3	4	4	4	2	3.00
Q.Avg	2.57	3.57	2.57	2.71	3.86	2.57	3.29	3.43	3.14	3.08[1]	3.00	3.57	2.57	2.57	3.86	2.43	3.29	3.43	3.57	3.14[1]

[1] These values refer to the Q.Avg values, i.e., the average of the rows to the left of it in either the pre or post-workshop columns.

Table 12: Comparison table of participant and question averages

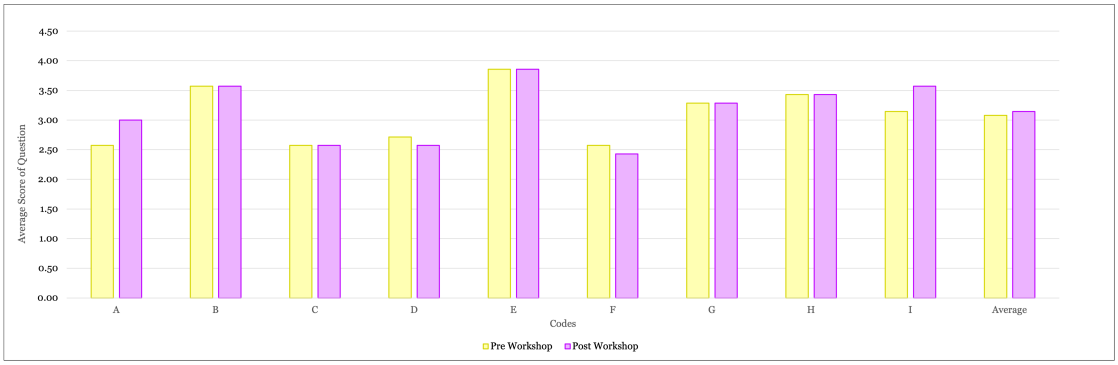


Figure 8: Comparison of question averages pre and post-workshop.

D.4 Sample Label Applications

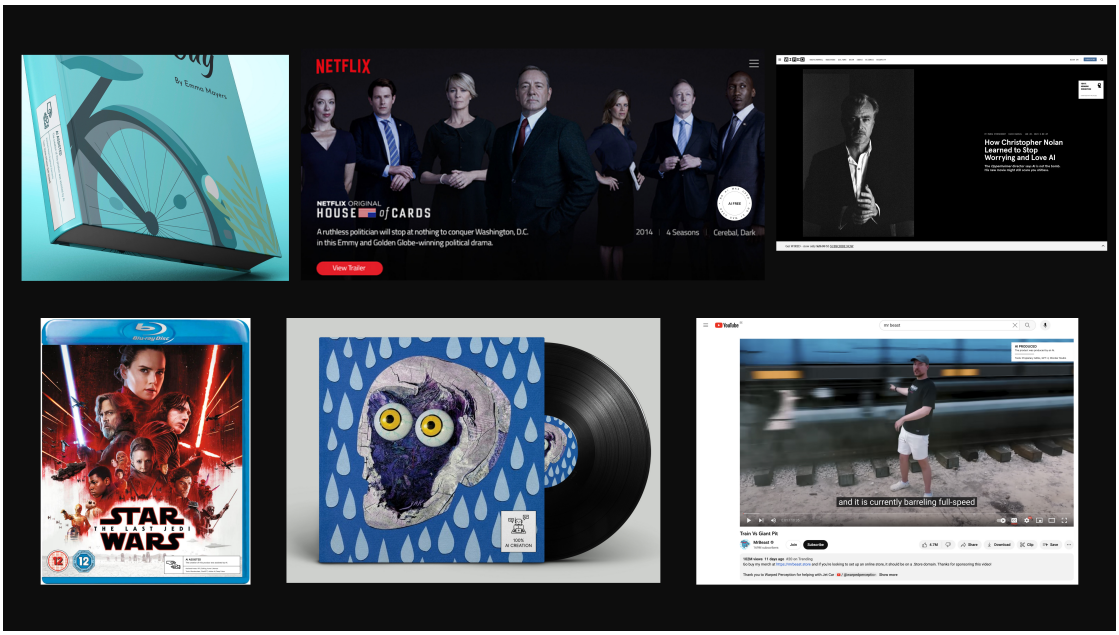


Figure 9: Sample label applications of pre-designed labels presented to participants.

E ENDNOTES

(1) The pronouns "they" and "their" are used as gender-neutral pronouns to refer to all participants/respondents in the third person out of respect for their gender identities as these were never asked or clarified.

(2) Workshop participant quotes and paraphrasing can be distinguished by the use of ("xyz") or lack of (xyz) quotation marks respectively.

(3) Quotes from workshop participants are provided after removing fillers such as "uh", "um", "like", or after removing parts of the quote that may contain personal information. Further, the quotes have been edited to suit the written format, as people speak differently from writing. This also accounts for participants who did not speak English as a first language and had trouble communicating certain sentences. Overall, care has been taken to ensure that the meanings of the statements are not altered.

(4) "War in Ukraine" refers to the 2022 Russian invasion of Ukraine [1].

(5) Decimal values are truncated to the nearest 2 decimal places after rounding using the rule of rounding up if the face value of the decimal place is equal to or greater than 5.

(6) The word "products" has been used in multiple places and still refers to artistic creations or other creative outputs, and is used simply due to direct quotations or semantic reasons.

F ATTRIBUTIONS OF ICONS USED

The following icons were sourced from The Noun Project: (1) human and robot hands by Berkah Icon, (2) Human in Love by NeueDeutsche, (3) AI by James Fok

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